

نظام تعليم البرمجة القائم على المشاريع

Project-Based Programming Education System

تقرير مشروع تخرج ١ - مقدم لاستكمال متطلبات الحصول على درجة البكالوريوس في هندسة المعلوماتية

إعداد الطلاب:

ملكة حرح

رغد ليلى

إشراف

م. أنس عبد العزيز

2025-2026

Abstract

In the era of rapid technological advancement and increasing reliance on digital solutions, online platforms have become vital tools for managing and enhancing business operations. This project presents a comprehensive digital system designed to [mention the main goal of your project, e.g., facilitate online interactions, improve service delivery, enhance user experience, etc.] for [mention the target users, e.g., students, professionals, local businesses, etc.].

The system aims to provide an intuitive, secure, and efficient platform that empowers users to [mention key functions, e.g., access resources, manage content, communicate, etc.] with ease and reliability. By leveraging modern technologies and user-centered design principles, the project seeks to address existing challenges and bridge gaps in [mention the domain, e.g., education, commerce, community services, etc.].

In this report, we outline the system's functional and non-functional requirements, architectural design, implementation details, and testing results. The platform is built using [mention technologies used, e.g., Django, Bootstrap, MySQL] and follows best practices in software engineering to ensure scalability, maintainability, and a seamless user experience.

We believe this system will serve as a valuable tool for [target users] and contribute positively to [mention broader impact, e.g., digital transformation, local economic support, enhanced accessibility, etc.].

ملخص

في عصر التقدم التقني المتسارع والاعتماد المتزايد على الحلول الرقمية، أصبحت المنصات

الإلكترونية أدوات حيوية لإدارة وتعزيز العمليات التجارية والخدمية. يقدم هذا المشروع نظاماً رقمياً شاملاً مصمماً لـ [اذكر الهدف الرئيسي لمشروعك، مثل: تسهيل التفاعلات عبر الإنترنت، تحسين تقديم الخدمات، تعزيز تجربة المستخدم، إلخ] من أجل [اذكر الفئة المستهدفة، مثل: الطلاب، المحترفين، الأعمال المحلية، إلخ].

يهدف النظام إلى توفير منصة سهلة الاستخدام وآمنة وفعالة تمكن المستخدمين من [اذكر الوظائف الرئيسية، مثل: الوصول إلى الموارد، إدارة المحتوى، التواصل، إلخ] بكل سهولة وموثوقية. من خلال الاستفادة من التقنيات الحديثة ومبادئ التصميم المركز على المستخدم، يسعى المشروع إلى معالجة التحديات الحالية وسد الفجوات في [اذكر المجال، مثل: التعليم، التجارة، الخدمات المجتمعية، إلخ].

في هذا التقرير، نقدم لمحة عن متطلبات النظام الوظيفية وغير الوظيفية، وتصميمه المعماري،

اذكر التقنيات المستخدمة، مثل [وتفاصيل التنفيذ، ونتائج الاختبارات. تم بناء المنصة باستخدام

وتتبع أفضل الممارسات في هندسة البرمجيات لضمان Django، Bootstrap، MySQL]

قابلية التوسع، وسهولة الصيانة، وتجربة مستخدم سلسلة

نؤمن أن هذا النظام سيكون أداة قيمة لـ [الفئة المستهدفة] وسيساهم إيجابياً في [اذكر الأثر الأوسع، مثل: التحول الرقمي، دعم الاقتصاد المحلي، تحسين إمكانية الوصول، إلخ]

References

المحتويات

CHAPTER 1 : INTRODUCTION	8
1.INTRODUCTION :	9
2. PROBLEM STATEMENT:	9
PROJECT OBJECTIVES :	10
PROPOSED SOLUTION :	10
PROJECT SCOPE:	11
PROJECT CONSTRAINTS:	12
PROJECT CONTRIBUTION :	12
RESEARCH METHODOLOGY :	13
REPORT STRUCTURE :	14
SUMMARY :	14
CHAPTER 2 : DEVELOPMENT AND PROJECT MANAGEMENT METHODOLOGY	15
2.1 SELECTED DEVELOPMENT METHODOLOGY AND JUSTIFICATIONS :	16
2.2 GANTT CHART PROJECT PLAN :	16
2.3 RISK MANAGEMENT TABLE :	17
2.4 THE TEAM IS DIVERSE AND THE ROLES ARE VARIED.	18
2.5 QUALITY MANAGEMENT :	18
2.6 SUMMARY :	18
CHAPTER 3 : REVIEW OF PREVIOUS LITERATURE AND STUDIES	19
3.1 THE REFERENCE STUDY :	20
3.2 MODERN TECHNIQUES AND FRAMWORKS USED :	20
3.3 RESEARCH GAP AND PROPOSED INNOVATION :	20
3.4 PROPOSED INNOVATION :	21
3.5 SUMMARY :	21
CHAPTER 4 : REQUIREMENTS MODELLING AND ANALYSIS	22
4.1 FEASIBILITY STUDY :	23
4.2 FUNCTIONAL REQUIREMENTS :	23
4.2.1 Account Management :	23
4.2.2 Learning Path Management :	24
4.2.3 Project Management :	24
4.2.4 Dashboard :	27
4.3 NON FUNCTIONAL REQUIREMENTS :	27
4.4 UML SCHEMES FOR REQUIREMENTS :	28
4.4.2 System features – use case specifications) Activity diagrams -Sequence diagrams (:	28
MODELING ACCORDING TO THE SELECTED METHODOLOGY :	40
Increment 1: Core System Features :	41
Increment 2: Advanced Learning and Project Features:	41
4.6 SUMMARY :	42
4.8 INITIAL TEST CASES :	45
4.8 SUMMARY :	54

CHAPTER 5 : ARCHITECTURAL DESIGN OF THE SYSTEM	55
5.1 HIGH-LEVEL DESIGN :	56
5.1.1 <i>System Architecture Design</i> :	57
5.1 DATABASE DESIGN :	59
5.1.1 <i>Constraints and Keys</i> :	60
5.1 <i>External APIs Design</i> :	61
5.1 SUMMARY :	62
CHAPTER 6 : EXECUTION ENVIRONMENT AND TOOLS.....	63
6.1 TOOLS USED :	64
6.2 DEVELOPMENT ENVIRONMENT AND PROJECT STRUCTURE :	64
6.3 PROGRAMMING LANGUAGES AND FRAMEWORKS :	64
6.4 DATABASE SERVER AND OPERATING SYSTEM :	64
6.5 DEPENDENCIES :	64
6.6 VERSION CONTROL AND SOURCE CODE MANAGEMENT :	64
6.7 CONTINUOUS INTEGRATION AND DEPLOYMENT (CI/CD):	65
6.8 TESTING ENVIRONMENT :	65
6.9 SUMMARY :	65
CHAPTER 7 : PROGRAMMATIC IMPLEMENTATION	66
7.1 PROJECT STRUCTURE :	67
7.2 MODULES IMPLEMENTATION :	68
7.3 API IMPLEMENTATION :	69
8.3 TEST CASE DIAGRAMS :	81
8.3.1 <i>Test Case Diagrams</i> :	81
8.7 SUMMARY :	84
CHAPTER 9 : RESULTS AND ANALYSIS	85
9.1 SYSTEM IMPLEMENTATION RESULTS :	86
9.2 COMPARATIVE ANALYSIS WITH EXISTING SYSTEMS:	87
9.3 ACHIEVEMENT OF OBJECTIVES :	88
9.4 STRENGTHS AND WEAKNESSES ANALYSIS :	88
9.6 SUMMARY :	89
CHAPTER 10 : CONCLUSION AND RECOMMENDATIONS	90
10.1 PROJECT SUMMARY AND MAIN ACHIEVEMENTS :	91
10.2 CHALLENGES AND DIFFICULTIES:	91
10.3 FUTURE WORK:	92
10.4 RECOMMENDATIONS:	92
10.5 SUMMARY :	92

List of tables

Risk management.....	17
The reference study	20
Assessing the learners' level upon joining the course (UC-1):	29
Project Access Dependency Management (UC-2):	31
Using the built-in code editor and implementing the solution (UC-3):	33
Project evaluation (automatic and manual) (UC-4):	35
Request for hints (UC-5):.....	37
View statistics and modification history (UC-6):	38
4.7 Complete Requirements Traceability Matrix (RTM) :	42
Sce-01: Login functionalities	45
Sce-02: Create account functionalities	46
Sce-03: Profile Management functionalities.....	46
Sce-04: Add Learning Path functionalities	47
Sce-05: Edit Learning Path functionalities.....	48
Sce-06: Delete Learning Path functionalities	48
Sce-07: View Learning Path functionalities.....	48
Sce-08: Join Learning Path functionalities	49
Sce-09: Learner Level Assessment functionalities	49
Sce-10: Add Project functionalities	50
Sce-11: Edit Project functionalities	50
Sce-12: Delete Project functionalities.....	51
Sce-13: View Projects functionalities	51
Sce-14: Start Project functionalities.....	51
Sce-15: Code Editor and Execution functionalities	52
Sce-16: Hints functionalities	52
Sce-17: Project Submission and Evaluation functionalities	53
Sce-18: Dashboard functionalities	53
Sce-04: Add Learning Path functionalities	81
Sce-15: Code Editor and Execution functionalities	82
Sce-16: Hints functionalities	82
8.3.2 Test Results :.....	83
8.4 Test Results and Analysis :	83

List of Figures

Gantt chart	16
4.4.1 Use Case Diagrams :	28
Activity diagram :	30
Sequence diagram :	30
Activity diagram :	31
Sequence diagram :	32
Activity diagram :	33
Sequence diagram :	34
Activity diagram :	35
Sequence diagram :	36
Activity diagram :	37
Sequence diagram :	38
Activity Diagram :	39
Sequence diagram :	40
5.1.1 System Block Diagram :	56
5.1.1Architectural design style used :	57
5.1.1 ER Diagram :	59
5.1.1 Relational Schema :	60
Learner Dashboard :	74
Learning Paths Page :	75
Project Details Page :	76
Code Editor Page :	76
Instructor Dashboard / Project Management Page :	77
Login page :	86
Code Editor and Execution Page :	87

Chapter 1 : Introduction

1.Introduction :

This chapter introduces the project by discussing the core problems and motivations behind developing this system. It outlines the project's aims and objectives, and provides a comprehensive overview of the report's structure.

2. Problem Statement:

Programming learners often struggle with traditional curricula that focus heavily on theoretical knowledge while providing simplified examples disconnected from real-world applications. This results in difficulty applying theoretical concepts to practical projects, leading to a gap in professional readiness for the digital job market. Furthermore, many online platforms lack a structured integration between theory and practice, failing to offer clear, systematic learning paths from beginner to advanced levels. Consequently, there is an urgent need for an innovative educational system that merges theoretical understanding with practical application through tiered, real-world projects.

Project Objectives :

The primary goal of this project is to develop an interactive "Project-Based Learning" (PBL) platform that enhances programming education within a structured digital environment. The specific objectives include:

- Establishing clear, logical learning paths for students to progress from beginner to advanced stages
- Providing self-assessment tools and immediate feedback to improve programming performance.
- Bridging the gap between theory and practice to prepare learners for professional software development roles.

Proposed Solution :

The proposed system is an interactive programming education platform centered on practical projects. By offering a regulated digital environment with projects of increasing difficulty, the system ensures that knowledge is transformed into actual skills, effectively qualifying learners for the labor market.

Project Scope:

The scope of this project involves the development of a comprehensive web-based platform tailored for programming education. The system includes:

- **User Management:** A secure authentication system for students and instructors.
- **Learning Paths:** Structured modules categorized by difficulty levels (Beginner, Intermediate, Advanced).
- **Project Repository:** A curated collection of real-world projects with theoretical support and technical requirements.
- **Progress Tracking:** Tools for students to monitor their completion rates and for instructors to assess performance.
- **Interactive Dashboard:** A user-centric interface for managing tasks, viewing feedback, and accessing resources.
- **Technology Stack:** Implementation using **Django** for the backend and **React** for the frontend, ensuring a scalable and responsive architecture.

Project Constraints:

The development and implementation of the project are subject to the following constraints:

- **Time Constraint:** The project must be completed, tested, and documented within the academic semester timeframe according to the university's graduation schedule.
- **Technical Constraints:** The system must be optimized to run on modern web browsers and requires a stable internet connection for real-time database interactions.
- **Resource Constraints:** Development is limited to the available hardware and free-tier cloud services for deployment (if applicable).
- **Language:** The platform's primary interface and educational content are currently limited to Arabic.

Project Contribution :

- The project provides several innovative contributions to the field of educational technology (EdTech) and software engineering training:
- Integration of Theory and Practice (Hybrid Learning Model): Unlike traditional platforms that separate theoretical lessons from coding exercises, this project introduces a unified environment where learners apply concepts directly to structured projects. It bridges the gap between "knowing" and "doing" by simulating a professional development workflow.
- Tiered Project-Based Curriculum: The project contributes a novel organizational structure for programming curricula, categorizing content

into "Learning Paths" with incremental difficulty levels. This approach ensures a smoother learning curve and helps students build a professional portfolio as they progress.

□ Centralized Resource Management for Instructors: The project provides a comprehensive management dashboard that allows instructors to monitor multiple students' progress and manage diverse learning paths in one place, which is a significant improvement over fragmented learning methods.

Research Methodology :

This research follows a hybrid scientific approach to ensure the academic accuracy of the report and the technical efficiency of the developed system:

- The Engineering Method: This is the primary methodology used for the development and documentation of the "Buildly" platform. It follows a systematic cycle that starts with problem identification, followed by requirement analysis, system design, implementation, and rigorous testing. This method ensures that the final product is a functional, scalable, and reliable engineering solution that meets the predefined educational goals.
- The Descriptive Analytical Method: This method is employed to study the current state of programming education and analyze existing platforms. By describing the gap between theoretical knowledge and practical application, this approach provides the necessary justification for the project's features. It involves analyzing the functional requirements

(FRs) and non-functional requirements (NFRs) to ensure the system effectively addresses the identified problem.

- **Comparative Analysis:** Used during the literature review phase to compare different educational models and technologies. This helps in selecting the most appropriate tools (such as Django and React) and pedagogical strategies (Project-Based Learning) to enhance the learning experience.

Report Structure :

The report is organized as follows:

- **Chapter 1:** Introduction.
- **Chapter 2:** Basic Concepts and Literature Review.
- **Chapter 3:** Project Management.
- **Chapter 4:** System Analysis.
- **Chapter 5:** System Design.
- **Chapter 6:** Implementation and Testing.
- **Chapter 7:** Conclusion and Future Work.

Summary :

This chapter establishes the need for a project-based programming learning system, explains the project objectives and proposed solutions, and provides an overview of the project structure. These scenarios pave the way for engaging with each aspect of the project in subsequent chapters.

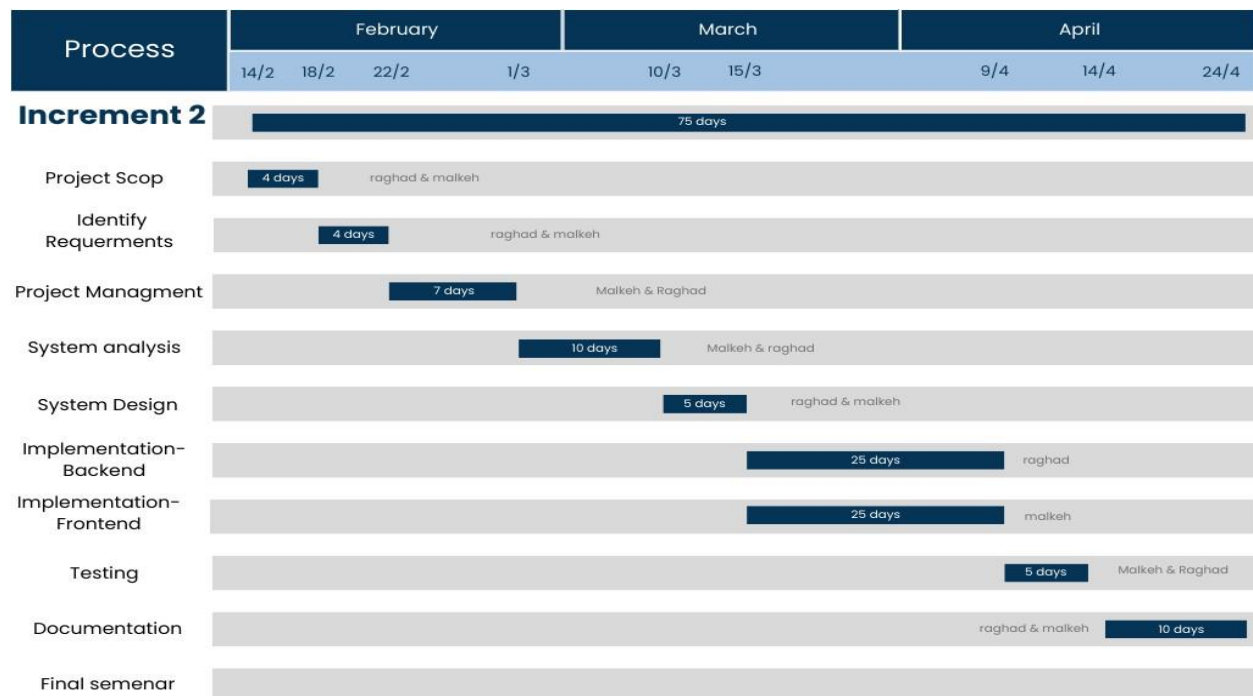
Chapter 2 : Development and Project Management Methodology

2.1 Selected Development Methodology and Justifications :

The selected development methodology for this project is the Incremental Development Methodology. This methodology was chosen because the Project-Based Programming Education System includes several modules and features that can be developed, tested, and improved gradually. Instead of building the entire system at once, the project is divided into smaller increments, where each increment delivers a functional part of the system.

2.2 Gantt Chart project plan :

Gantt chart



2.3 Risk Management Table :

Risk management

Risk	Likelihood	Impact	Mitigation plan
Schedule Delay	High	High	Divide the project into smaller increments, prioritize tasks, and follow a clear timeline using planning tools such as Gantt Chart.
Difficulty in learning new technologies	Medium	High	Allocate time for self-learning, use online tutorials and documentation, and conduct regular practice sessions.
Integration issues	Medium	High	Perform continuous integration and frequent testing, maintain clear API documentation, and ensure consistent communication between components.
Limited testing resources	Medium	Medium	Apply manual testing for each feature, write test cases, and validate each increment before moving to the next phase.
Time constraints	High	High	Focus on core features first, manage time efficiently, and postpone non-essential features if necessary.
Technical complexity	Medium	High	Implement features incrementally, start with basic functionality, and gradually enhance complexity with testing at each stage.

2.4 The Team Is Diverse And The Roles Are Varied.

Stakeholder	Role	Responsibility
Eng. Anas Abdelaziz	Project Manager & Supervisor	Guides the project and provides feedback.
Raghad Laila	Software Engineer	Develop backend functionalities, implement authentication, handle database integrations and documentation.
Malakeh Herhe	Software Engineer	Frontend development, ensure a seamless user experience and documentation.

2.5 Quality Management :

Quality management is an essential aspect of the software development process, as it ensures that the final system meets the required standards and satisfies user expectations. In this project, several practices and strategies are adopted to maintain and improve the quality of the Project-Based Programming Education System throughout its development lifecycle.

2.6 Summary :

This chapter presented the selected development methodology, project planning, risk management, team roles, and quality management practices. The Incremental Development Methodology was adopted to support gradual system development and testing. These management and planning activities helped organize the development process and ensure project quality.

Chapter 3 : Review of previous literature and studies

3.1 The reference study :

The reference study

feature	Codecademy	Freecodecamp	SoloLearn	The Odin Project	النظام المقترح
التعلم القائم على المشاريع	X	✓	X	✓	✓
محرك أكواد مدمج	✓	✓	✓	X	✓
تشغيل الكود وتنفيذه	✓	✓	✓		✓
لوحة تحكم واحصائيات مفصلة لتقدم المتعلم	✓	✓	✓	✓	✓
تحديد تبعيات المشاريع	✓	X	✓	X	✓
مشاريع تعليمية منظمة ومتدرجة	✓	✓	✓	✓	✓
إرفاق ملفات بداية تحتوي على هيكل الكود	✓	✓	X	✓	✓
واجهة تفاعلية سهلة الاستخدام	✓	✓	✓	✓	✓
مشاريع متدرجة الصعوبة ضمن المسار التعليمي	X	✓	X	✓	✓
تقييم مستوى المتعلم عند الانضمام للمسار	X	X	X	X	✓
تخصيص المشاريع حسب مستوى المتعلم	X	X	X	X	✓
حفظ حالة المشروع تلقائياً	X	X	X	X	✓
تقديم يدوي من المشرف للمشاريع	X	X	X	X	✓
سجل تعديلات والرجوع لإصدار سابق	X	X	X	X	✓
تلميحات تدريجية hints داخل المشروع	X	X	X	X	✓

3.2 Modern techniques and frameworks used :

Programming languages : Css , Html , Javascript .

Framework : React(frontend) , Django(backend) .

DataBase : PostgreSQL .

3.3 Research Gap and Proposed Innovation :

Based on the analysis of existing platforms such as Codecademy, FreeCodeCamp, SoloLearn, and The Odin Project, several limitations are identified. Most platforms do not provide a fully integrated project-based learning experience, and they lack adaptive learning that adjusts content according to the learner's level. Additionally,

features such as project dependencies, detailed progress tracking, and manual evaluation with personalized feedback are either missing or not well implemented.

3.4 Proposed Innovation :

The proposed system addresses these gaps by introducing an integrated platform that combines project-based learning, initial level assessment, structured learning paths with dependencies, and a built-in code editor. It also provides automatic and manual evaluation, progress tracking, and incremental hints, offering a more effective and practical learning experience.

3.5 Summary :

This chapter reviewed previous studies and existing programming learning platforms. It also discussed the technologies and frameworks used in the project and highlighted the research gap addressed by the proposed system. The chapter demonstrated how the project provides a more practical and integrated projectbased learning experience.

Chapter 4 : Requirements Modelling And Analysis

4.1 Feasibility Study :

The feasibility study evaluates the possibility of successfully developing the proposed system.

From a technical perspective, the system can be implemented using technologies such as Django, React, and PostgreSQL, which are suitable and available.

From an operational perspective, the system is designed to be user-friendly and easy to use for both learners and instructors.

From a time perspective, the project can be completed within the academic semester by following the incremental development approach.

Overall, the project is feasible and can be successfully implemented within the available resources and constraints.

4.2 Functional Requirements :

4.2.1 Account Management :

FR-1.1: The system must allow the user (learner, trainer) to create a new user account in the system.

FR-1.2: The system must allow the user (learner, trainer) to log in using a username and password.

FR-1.3: The system must allow the user to view and manage profile information.

FR-1.4: The system should allow the user to edit profile information.

FR-1.5: The system must allow the user (learner, trainer) to log out of their account.

4.2.2 Learning Path Management :

FR-2.1 : The system must allow the instructor to create a new learning path, specifying its title, description, and content.

FR-2.2 : The system should allow the instructor to modify an existing learning path.

FR-2.3 : The system should allow the instructor to delete a learning path.

FR-2.4 : The system should display a list of available learning paths to the user.

FR-2.5 : The system must allow the learner to view information about a specific path.

FR-2.6 : The system must allow the learner to join an educational path.

FR-2.7 : The system should display the projects associated with each learning path to the learner when entering a path.

FR-2.8 : The system must assess the learner upon joining the course to determine their level (beginner, intermediate, advanced) and present them with suitable projects according to this level before starting any project.

4.2.3 Project Management :

FR-3.1 : The system should allow the trainer to add a new project with its details specified (title, objectives, tasks, resources, requirements).

FR -3.2 : The system should allow the coach to modify an existing project.

FR-3.3 : The system should allow the coach to delete the project.

FR-3.4 : The system must allow the user (administrator, learner) to view the list of projects in the system.

FR-3.5 : The system must allow the user (administrator, learner) to view the list of projects in a specific location.

FR-3.6 : The system must allow the administrator to view specific project information.

FR-3.7 : The system should allow the user to select a project and view educational project information.

FR-3.8 : The system should allow the learner to start a specific project.

FR-3.9 : The system should allow projects to be displayed according to their status (new, in progress, completed).

FR-3.10 : The system must allow the addition of a multi-stage project (sequential stages).

FR-3.11 : The system should allow for the definition of project dependencies so that the learner does not start a project until a previous project has been completed.

FR-3.12 : The system should allow projects to be filtered by level.

FR-3.13 : The system should allow the attachment of starter files to each project containing a preliminary code structure.

FR-3.14 : The system should allow the supervisor to "define" the prerequisites for each project when it is created.

FR-3.15 : The system should provide a built-in code editor with syntax highlighting for projects.

FR-3.16 : The system must support the execution of the code within the platform and the display of results.

FR-3.17 : The system should allow the project status to be saved automatically at specified time intervals.

FR-3.18 : The system should provide a progress bar that shows the percentage of project completion.

FR-3.19 : The system must support the display of detailed statistics on start time, completion time, and project time rate.

FR-3.20 : The system must support displaying the administrator's modification history with the ability to revert to a previous version.

FR-3.21 : The system should allow the learner to submit the project.

FR-3.22: The system should allow deleting tasks in the project.

FR-3.23 : The system should allow adding tasks during project modification.

FR-3.24 : The system must support providing incremental hints within the project without revealing the complete solution.

FR-3.25 : When a project is being executed, the system should display a list of task titles in the sidebar, highlighting the current task.

FR-3.26 : The system should allow the supervisor to manually evaluate the project and add text comments or a star rating.

FR-3.27 : The system should display a list of delivered projects to the administrator.

FR-3.28 : The system should filter submitted projects based on whether they have been evaluated or not.

4.2.4 Dashboard :

FR-4.1 : The system provides the learner with a control panel that displays the learning paths the learner has enrolled in, the number of projects being implemented and completed in the path, and the number of learning hours.

4.3 Non Functional Requirements :

NFR-4.3.1 : The system must encrypt all passwords using secure hashing algorithms and not save them as plain text to protect user data.

NFR-4.3.2 : The system should load any web page (except code editor pages) within a maximum of three seconds for 95% of users when using a normal internet connection.

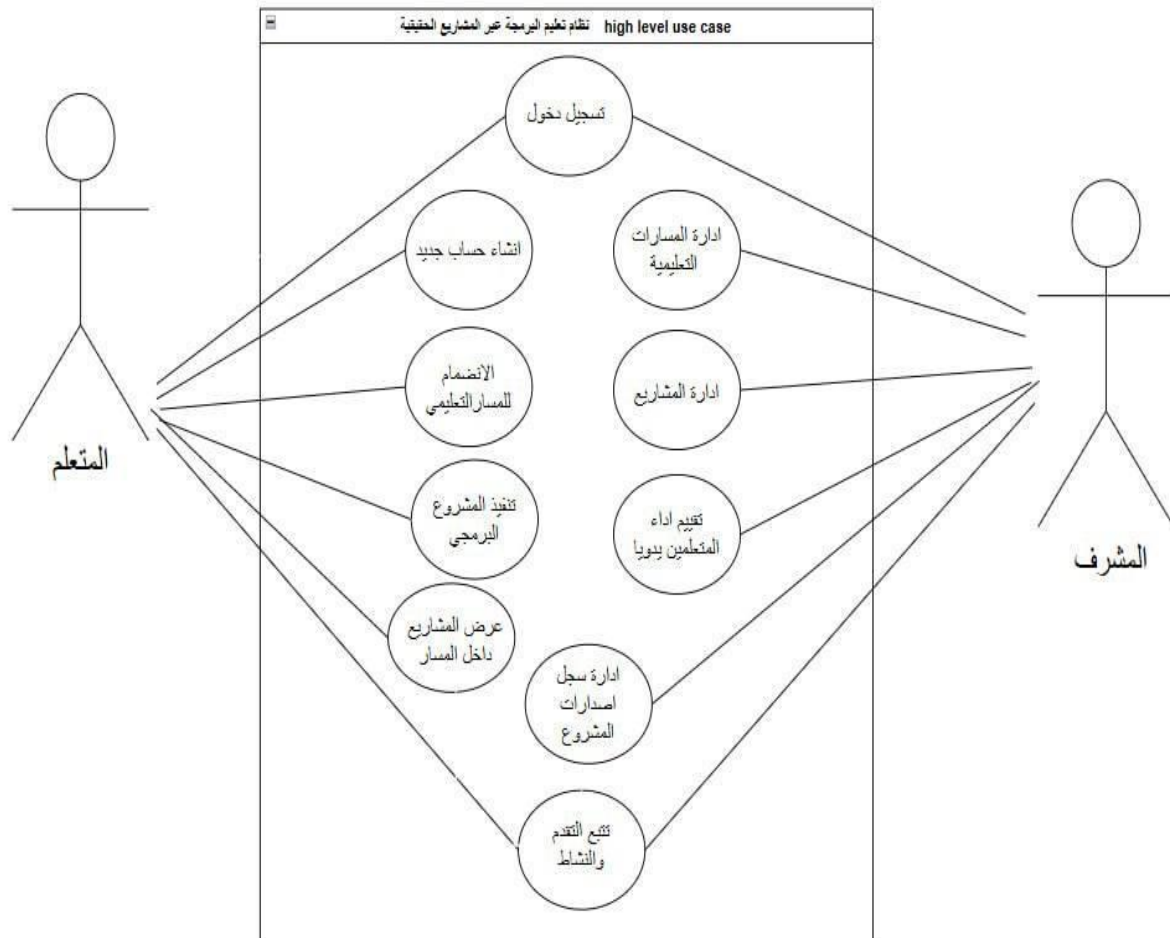
NFR-4.3.3 : The system must be scalable to support an increasing number of users without a noticeable drop in performance or the need for major infrastructure changes.

NFR-4.3.4 : The system should provide user-friendly interfaces so that a new user needs, on average, "10 minutes" to fully navigate and understand all parts and pages.

NFR-4.3.5 : The system should allow the addition of new features or functions with minimal changes to the existing codebase and without affecting the communication of core components with other new services.

4.4 UML Schemes for Requirements :

4.4.1 Use Case Diagrams :



4.4.2 System features – use case specifications) Activity diagrams - Sequence diagrams (:

Test Case Descriptions :

Introduction :

Test cases are used to verify that the system functions correctly according to the defined requirements. They help ensure that each feature operates as expected under different conditions.

In this project, a set of representative test cases is defined to validate the core functionalities of the system, such as account management, learning

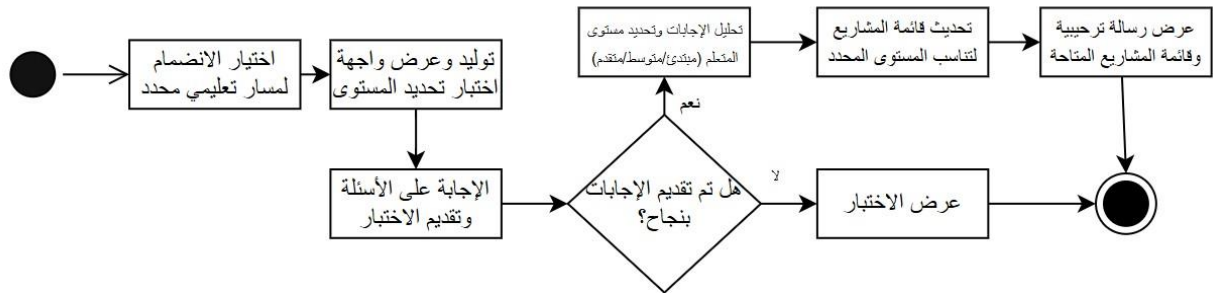
path operations, and project handling. These test cases cover the main user interactions and typical scenarios.

Due to the scope of the project, only a subset of test cases is presented in this report, focusing on the most important and frequently used features. Additional test cases can be defined and executed during further development and testing phases.

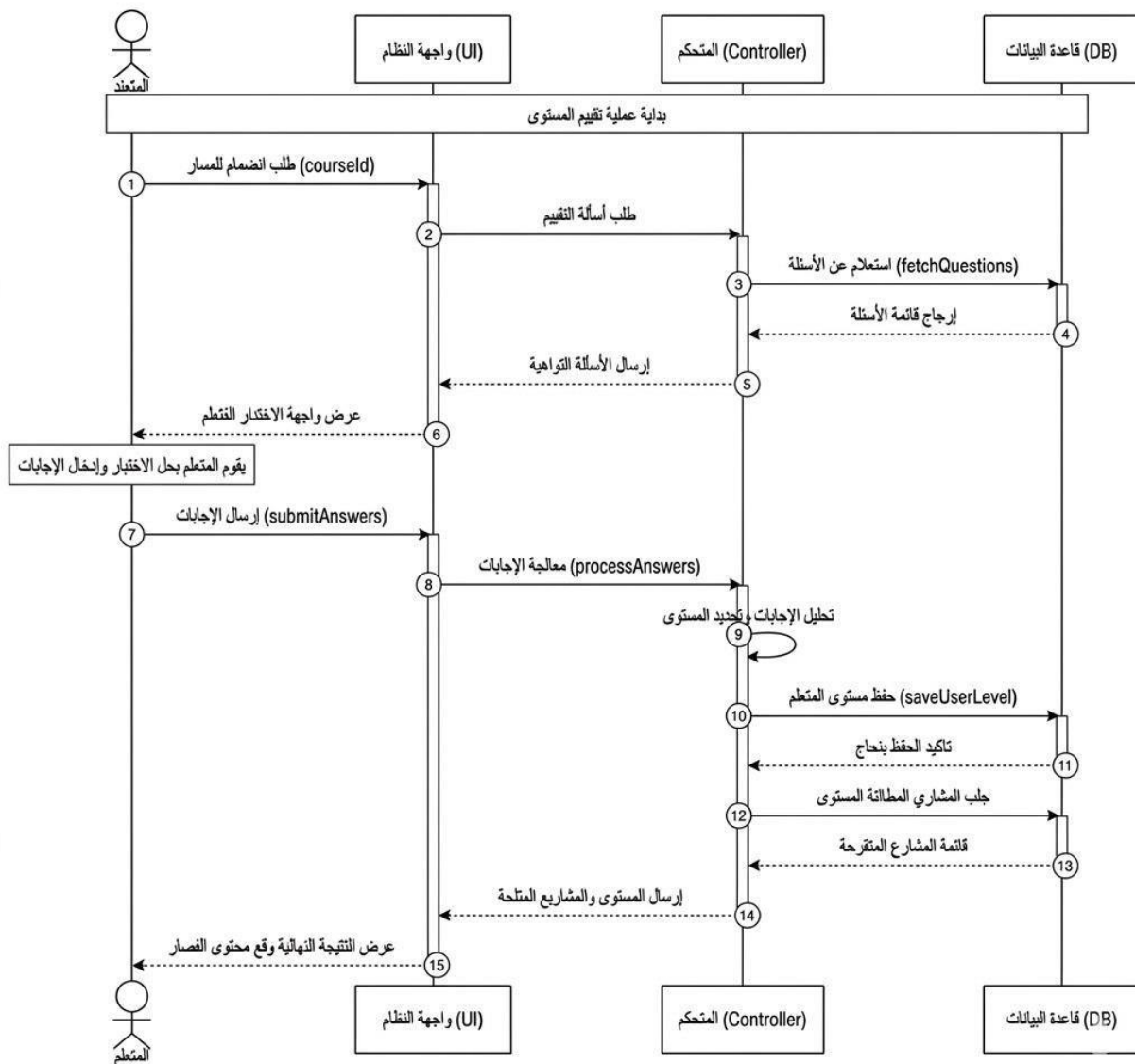
Assessing the learners' level upon joining the course (UC-1):

رقم الحالة	UC-1
اسم الحالة	تقييم مستوى المتعلم عند الانضمام للمسار
الفاعلون	المتعلم , النظام
الوصف	يقوم النظام بإجراء اختبار تقييمي للمتعلم فور انضمامه لمسار تعليمي لتحديد مستواه (مبتدئ , متوسط , متقدم) و عرض المشاريع المناسبة له.
الشروط المسبقة	1. يجب ان يكون المتعلم قد سجل دخوله للنظام يجب أن يقوم 2. المتعلم بالضغط على "انضمام" لمسار محدد.
التدفق الرئيسي	1. يختار المتعلم الانضمام لمسار تعليمي معين 2. يعرض النظام واجهة اختبار تحديد المستوى الخاصة بهذا المسار 3. يقوم المتعلم بالإجابة على الأسئلة و تقديم الاختبار يقوم النظام 4. بتحليل الإجابات و حساب النتيجة 5. يحدد النظام مستوى المتعلم (مبتدئ , متوسط , متقدم) 6. بعرض النظام رسالة ترحيبية و تظهر قائمة المشاريع المناسبة لمستوى المتعلم داخل المسار
التدفق البديل	[A1] . المتعلم يقرر عدم إجراء الاختبار الآن : ١. يغلق المتعلم واجهة الاختبار. ٢. يرسل النظام تنبيهها بان الوصول للمشاريع يتطلب تحديد المستوى أولاً.
الشروط اللاحقة	يتم تخزين مستوى المتعلم في قاعدة البيانات و تحديث قائمة المشاريع المتاحة له في لوحة تحكمه.

Activity diagram :



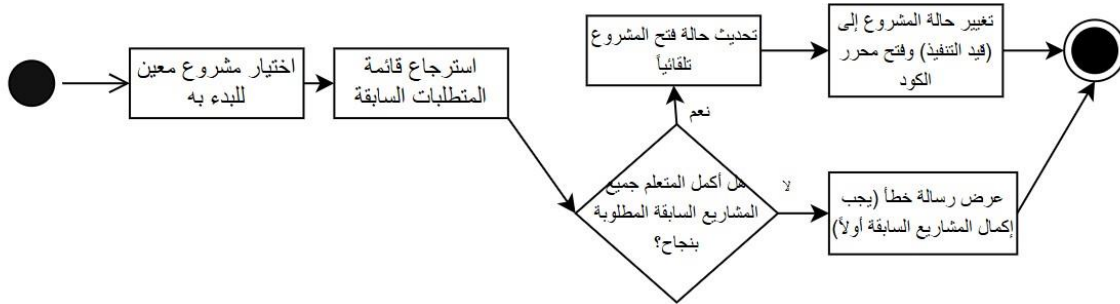
Sequence diagram :



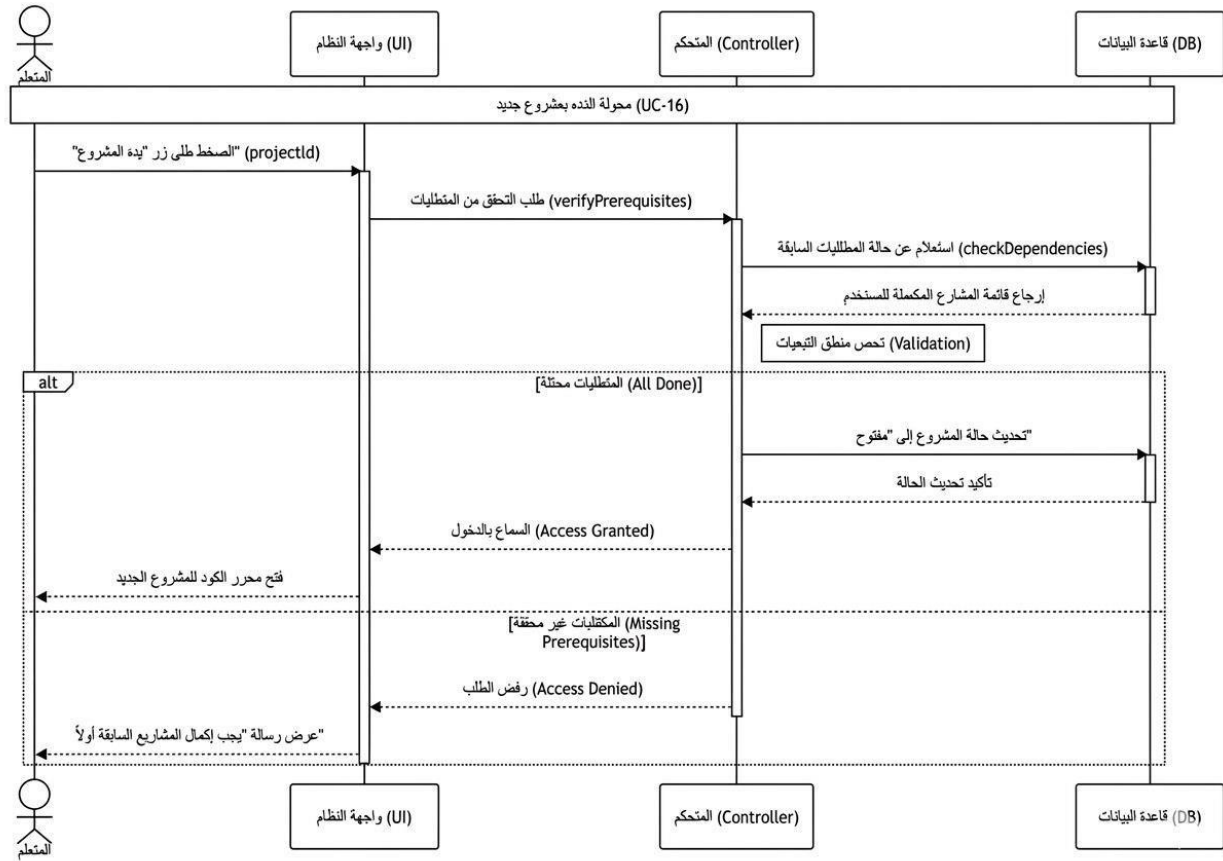
Project Access Dependency Management (UC-2):

رقم الحالة	UC-2
اسم الحالة	إدارة تبعيات الوصول للمشاريع
الفاعلون	المتعلم , النظام , المشرف
الوصف	تضمن هذه الحالة التزام المتعلم بتسلسل التعلم، حيث يمنع النظام الوصول لمشروع جديد إذا لم يتم إكمال المتطلبات السابقة المحددة من قبل المشرف.
الشروط المسبقة	1. وجود مشاريع منشأة مسبقاً في المسار. 2. قيام المشرف بتحديد المتطلبات السابقة لكل مشروع.
التدفق الرئيسي	1. يحاول المتعلم البدء بمشروع معين. 2. يقوم النظام بالتحقق من قائمة "المتطلبات السابقة" لهذا المشروع. 3. يتحقق النظام من حالة المشاريع السابقة للمتعلم (هل هي "مكتملة"؟). 4. إذا كانت جميع المتطلبات مكتملة، يفتح النظام المشروع للمتعلم. 5. يتم تحديث حالة المشروع إلى "قيد التنفيذ".
التدفق البديل	[A1] . المتطلبات السابقة غير مكتملة : 1. يظهر النظام رسالة للمتعلم: "يجب إكمال المشاريع [اسم المشروع] أولاً." 2. يبقى زر البدء بالمشروع غير فعال
الشروط اللاحقة	ضمان تسلسل العملية التعليمية وتحديث حالة "فتح المشاريع" تلقائي أ.

Activity diagram :



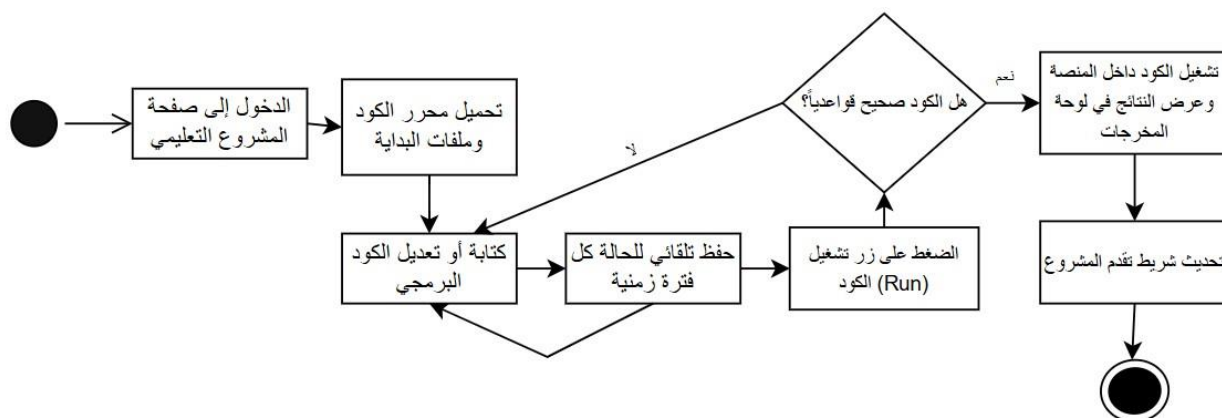
Sequence diagram :



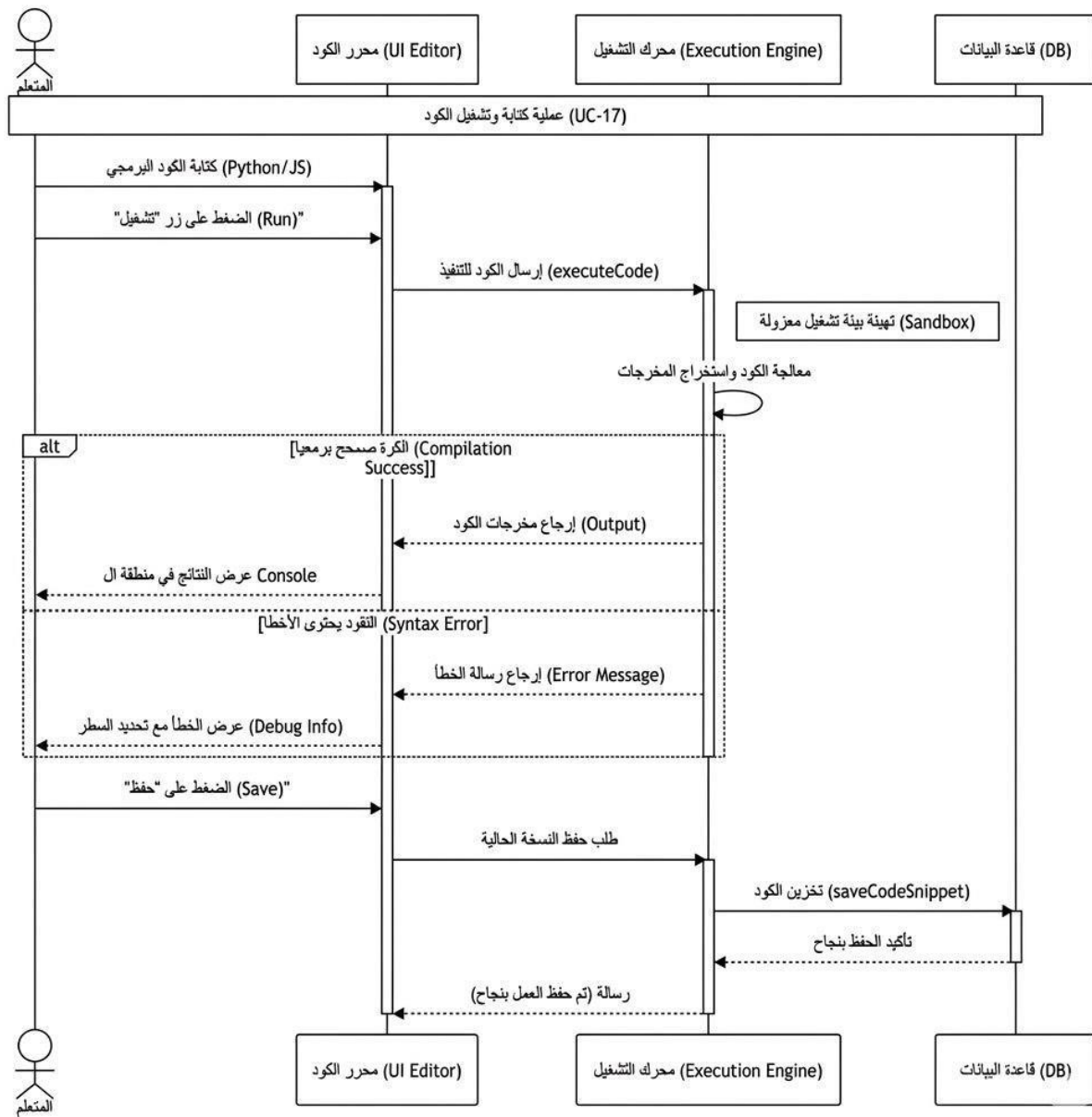
Using the built-in code editor and implementing the solution (UC-3):

UC-3	رقم الحالة
استخدام محرر الكود المدمج وتنفيذ الحل	اسم الحالة
المتعلم , النظام	الفاعلون
يوفر النظام بيئة تطوير متكاملة (IDE) داخل المتصفح تسمح للتعلم بكتابة الكود، تمييز بناء الجملة، تشغيل الكود، وحفظ الحالة تلقائياً.	الوصف
1. أن يكون المتعلم قد بدأ مشروعاً معيناً. 2. تحميل ملفات البداية (Starter Files) وهيكل الكود المبدئي.	الشروط المسبقة
1. يفتح المتعلم واجهة المشروع التعليمي. 2. يعرض النظام محرر الكود مع ملفات البداية المرفقة مسبقاً. 3. يقوم المتعلم بكتابة الكود البرمجي 4. يقوم النظام بحفظ حالة الكود تلقائياً كل فترة زمنية محددة. 5. يضغط المتعلم على زر "تشغيل". /Run 6. يقوم النظام بتنفيذ الكود في بيئة معزولة وعرض النتائج أو الأخطاء في لوحة المخرجات.	التدفق الرئيسي
[A1] . وجود خطأ برمجي : 1. يكتشف النظام الخطأ أثناء التشغيل. 2. يعرض النظام رسالة الخطأ للمتعلم في لوحة المخرجات مع تحديد رقم السطر إن أمكن.	التدفق البديل
تحديث سجل التعديلات الخاص بالمشروع لضمان إمكانية الرجوع لإصدارات سابقة	الشروط اللاحقة

Activity diagram :



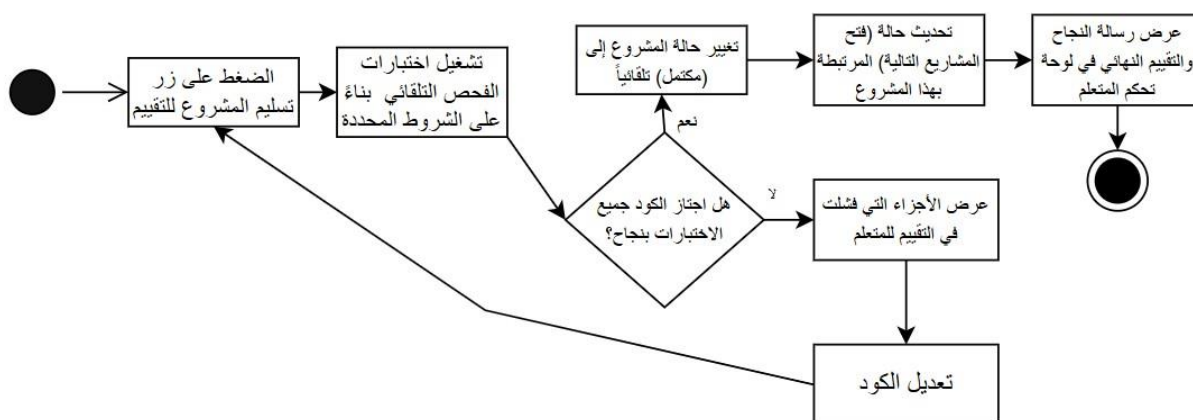
Sequence diagram :



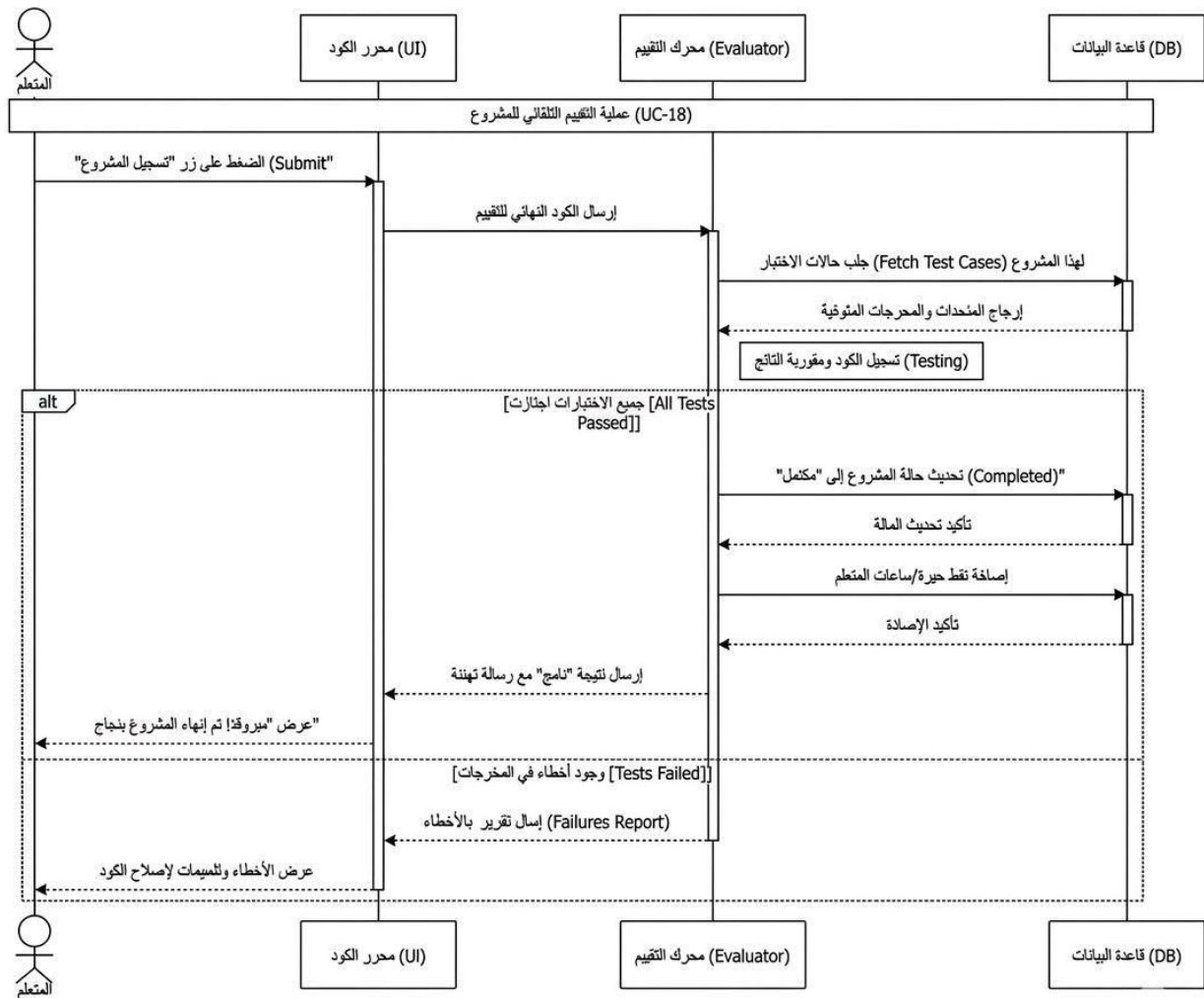
Project evaluation (automatic and manual) (UC-4):

رقم الحالة	UC-4
اسم الحالة	تقييم المشروع (تلقائي ويدوي)
الفاعلون	المتعلم , النظام , المشرف
الوصف	يتم تقييم عمل المتعلم برمجياً بناءً على شروط محددة مسبقاً، مع إمكانية تدخل المشرف للتقييم اليدوي وإضافة ملاحظات.
الشروط المسبقة	1. قيام المتعلم بإنهاء مهام المشروع والضغط على "تسليم". وجود 2. معايير تقييم (Test Cases) محددة عند إنشاء المشروع.
التدفق الرئيسي	1. يرسل المتعلم الكود النهائي للتقييم. 2. يقوم النظام بتشغيل الاختبارات التلقائية ومقارنة النتائج بالشروط المحددة. إذا نجح التقييم، يقوم النظام بتحديث حالة المشروع إلى "مكتمل" تلقائياً. 4. يقوم النظام بفتح المشاريع التالية التي كانت مقفلة (بناءً على التبعية).
التدفق البديل	[A1]. فشل التقييم التلقائي : 1. يظهر النظام للمتعلم الأجزاء التي لم تجتز الاختبار. 2. تظل حالة المشروع "قيد التنفيذ" حتى يتم التعديل.
الشروط اللاحقة	تحديث إحصائيات لوحة التحكم للمتعلم وظهور معدل التقدم الجديد.

Activity diagram :



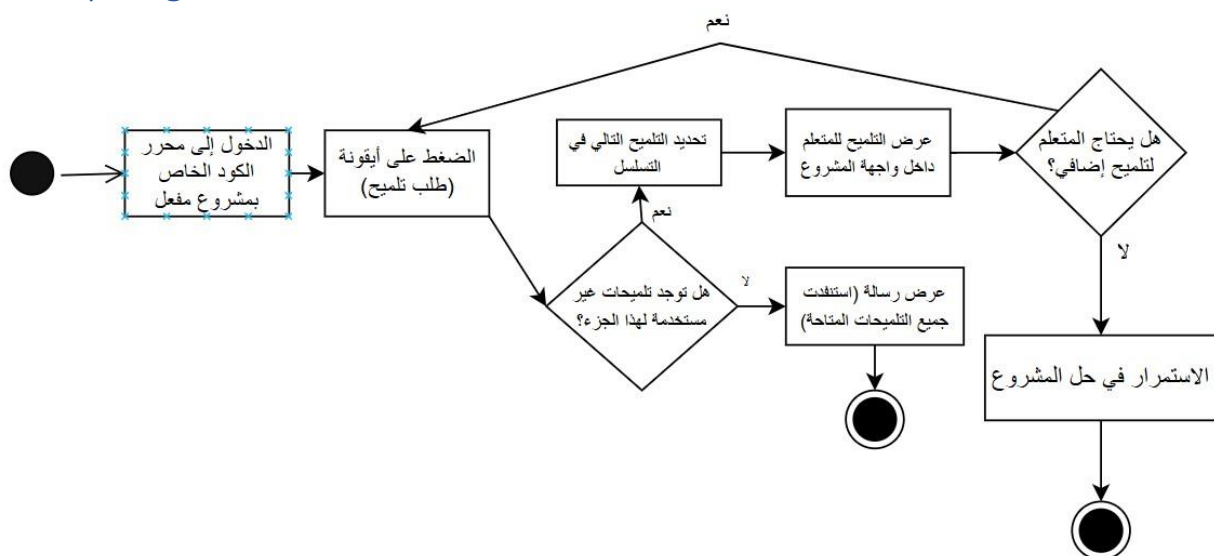
Sequence diagram :



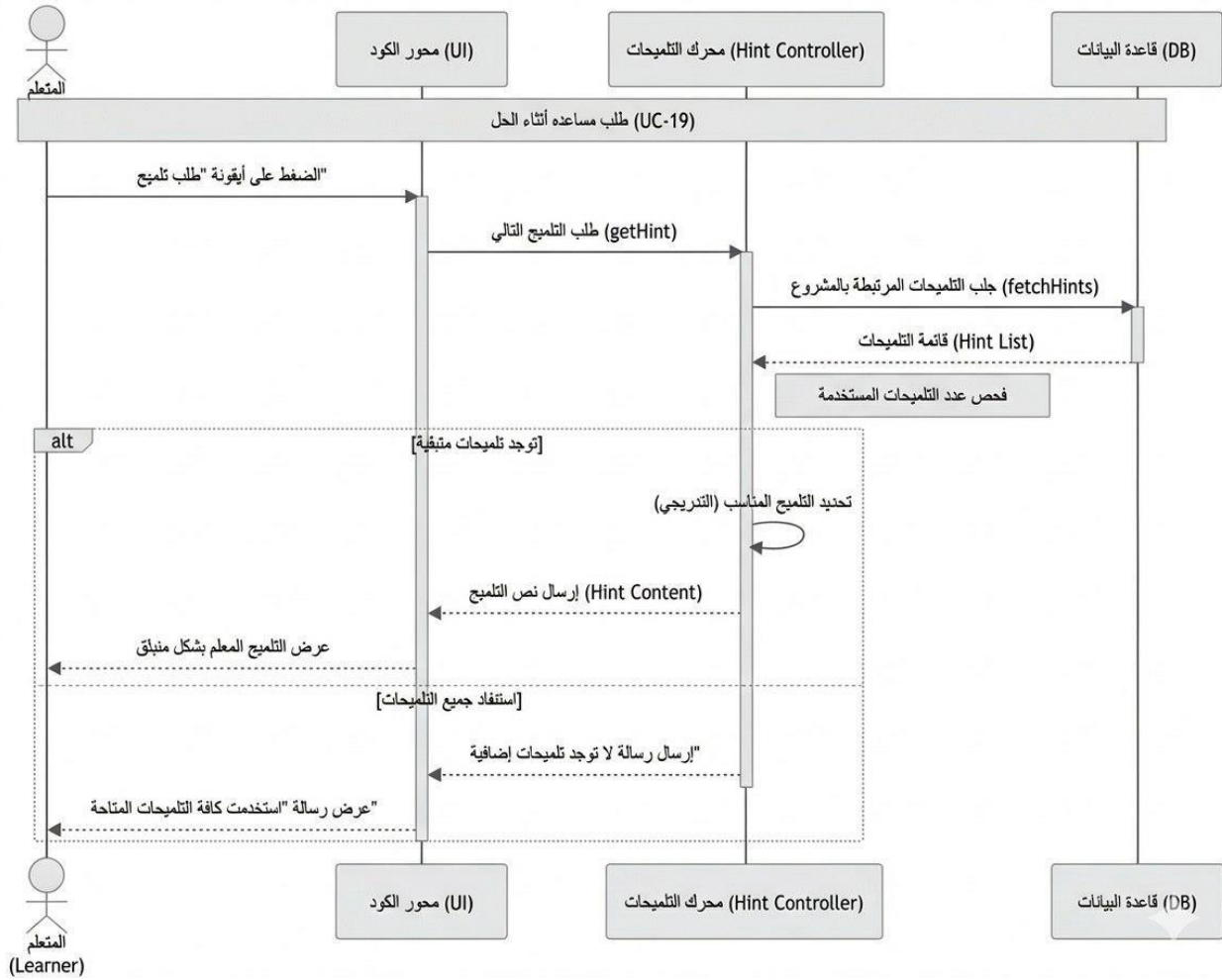
Request for hints (UC-5):

رقم الحالة	UC-5
اسم الحالة	طلب تلميحات (Hints)
الفاعلون	المتعلم , النظام
الوصف	يسمح النظام للتعلم بالحصول على مساعدة تدريجية أثناء حل المشروع دون كشف الحل الكامل مباشرة
الشروط المسبقة	1. أن يكون المتعلم داخل محرر الكود الخاص بمشروع مفعّل.
التدفق الرئيسي	١. يضغط المتعلم على أيقونة "طلب تلميح" ٢. يعرض النظام التلميح الأول (تلميح عام). ٣. إذا احتاج المتعلم للمزيد، يضغط مرة أخرى ليظهر تلميح أكثر تفصيلاً.
التدفق البديل	[A1] . نفاذ التلميحات:
	1. يعلم النظام المتعلم أنه قد استنفد جميع التلميحات المتاحة لهذا الجزء.
الشروط اللاحقة	قد يؤثر استخدام التلميحات على التقييم النهائي للمشروع (حسب إعدادات المشرف)

Activity diagram :



Sequence diagram :

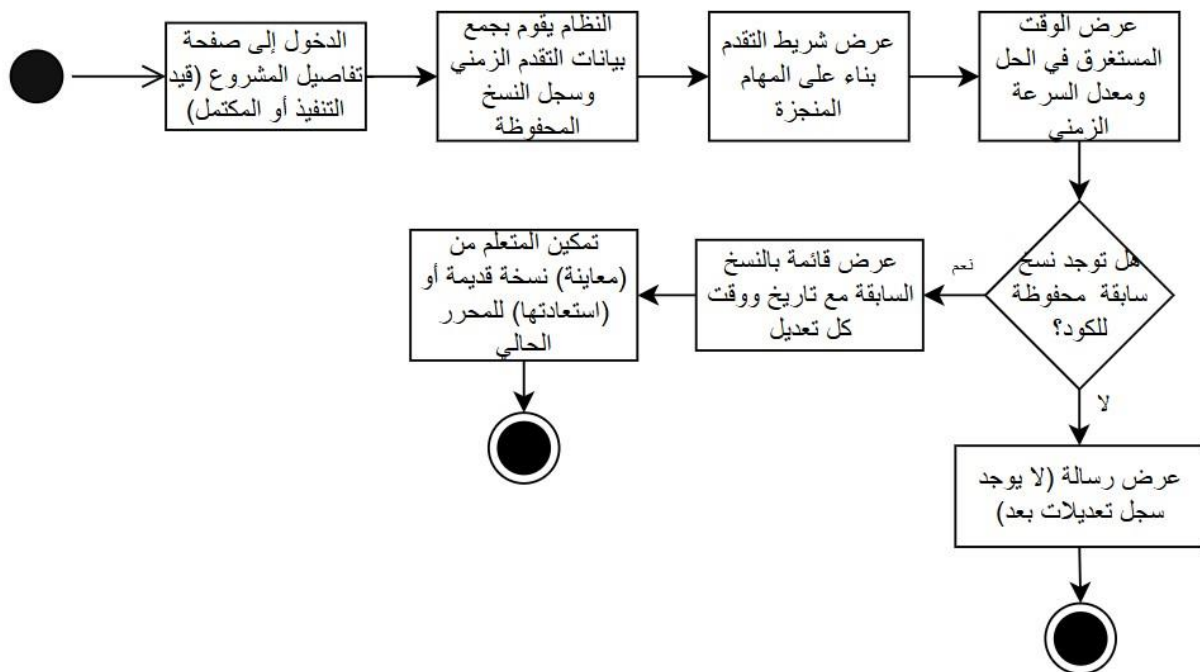


View statistics and modification history (UC-6):

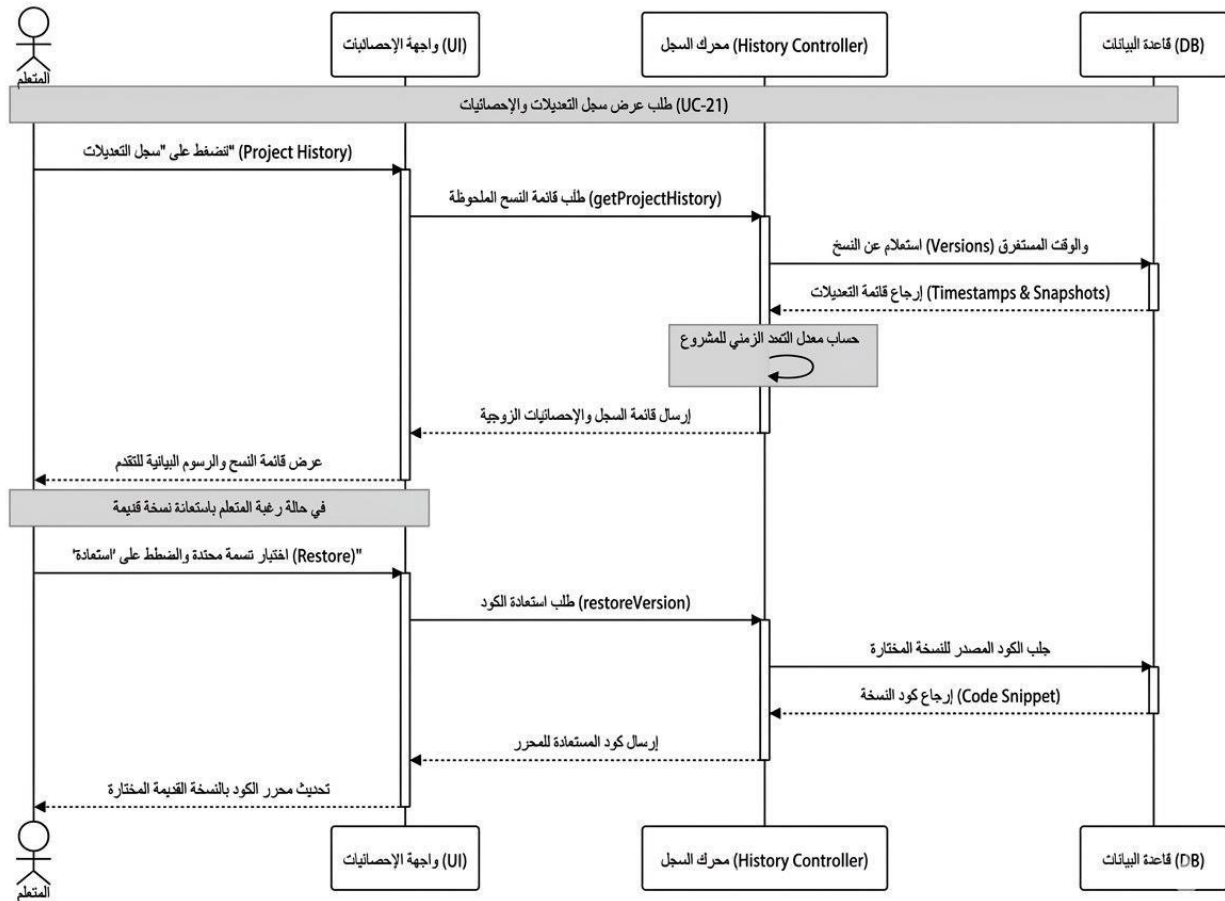
UC-6	رقم الحالة
عرض إحصائيات وسجل تعديلات المشروع	اسم الحالة
المتعلم , النظام	الفاعلون
يتيح النظام للمتعلم رؤية تفاصيل دقيقة عن أدائه داخل مشروع معين، بما في ذلك الوقت المستغرق وشريط التقدم وسجل النسخ السابقة.	الوصف
1. أن يكون المتعلم قد بدأ العمل على مشروع واحد على الأقل	الشروط المسبقة

التدفق الرئيسي	1. يفتح المتعلم صفحة تفاصيل المشروع الحالي. 2. يعرض النظام "شريط التقدم" الذي يوضح نسبة الإنجاز بناءً على المهام المنتهية. 3. يظهر النظام إحصائيات الوقت (وقت البدء، وقت الإنجاز، والمعدل الزمني). 4. يتيح النظام للمتعلم عرض "سجل التعديلات (History)" "الرؤية النسخ السابقة من الكود". 5. يمكن للمتعلم اختيار نسخة سابقة لاستعادتها إذا رغب في ذلك.
التدفق البديل	[A1] . لا يوجد سجل تعديلات:
	1. إذا لم يتم المتعلم بحفظ أي نسخة أو كان في بداية المشروع، يظهر النظام رسالة تفيد بعدم توفر سجلات بعد.
الشروط اللاحقة	تمكين المتعلم من إدارة وقته ومراجعة تطور الحل البرمجي الخاص به.

Activity Diagram :



Sequence diagram :



Modeling According to the Selected Methodology :

Since the project follows the Incremental Development Methodology, the system requirements were divided into feature lists according to development increments. Each increment includes a group of related functional requirements that can be developed, tested, and delivered as a working part of the system.

Increment 1: Core System Features :

The first increment focuses on the essential features required to make the system usable. It includes account management, learning path management, basic project management, and the learner dashboard.

Included features:

- A. Account creation.
- B. Login.
- C. Logout.
- D. profile management.
- E. Creating, editing, deleting, and viewing learning paths.
- F. Joining learning paths and viewing path details.
- G. Creating, editing, deleting, and viewing projects.
- H. Starting a project.
- I. Viewing the learner dashboard.

Increment 2: Advanced Learning and Project Features:

The second increment includes the advanced features highlighted in green in the requirements document. These features improve the learning experience and add more interaction and control within projects.

Included features:

- A. Learner level assessment when joining a learning path.
- B. Project dependencies and prerequisite management.

- C. Built-in code editor with syntax highlighting.
- D. Code execution inside the platform.
- E. Auto-saving project progress.
- F. Progress bar and time statistics.
- G. Version history and rollback.
- H. Incremental hints.
- I. Project submission and manual evaluation.
- J. Viewing and filtering submitted projects.

4.6 Summary :

This division helps organize the development process and reduce complexity. The first increment delivers the basic operational system, while the second increment enhances the platform with advanced project-based learning features.

4.7 Complete Requirements Traceability Matrix (RTM) :

Req_ID	Use Case	Title	Analysis	System Design	Coding	Test Cases
RE-FR-AM-01	UC-01	Create Account	Account creation	Authentication	Implemented	TC-04 , TC-05 , TC-06 , TC-07
RE-FR-AM-02	UC-02	Login	Login process	Authentication	Implemented	TC-01 , TC-02 , TC-03
RE-FR-AM-03	UC-03	View Profile	Profile view	Profile	Implemented	TC-08
RE-FR-AM-04	UC-04	Edit Profile	Profile edit	Profile	Implemented	TC-09 , TC-10 , TC-11

RE-FR-AM-05	UC-05	Logout	Logout process	Authentication	Implemented	—
RE-FR-LPM-06	UC-06	Create Path	Create path	Learning Path	Implemented	TC-12 , TC-13 , TC-14
RE-FR-LPM-07	UC-07	Edit Path	Edit path	Learning Path	Implemented	TC-15 , TC-16
RE-FR-LPM-08	UC-08	Delete Path	Delete path	Learning Path	Implemented	TC-17 , TC-18
RE-FR-LPM-09	UC-09	View Paths	View paths	Learning Path	Implemented	TC-19
RE-FR-LPM-10	UC-10	View Path Details	Path details	Learning Path	Implemented	TC-20
RE-FR-LPM-11	UC-11	Join Path	Join path	Learning Path	Implemented	TC-21 , TC-22
RE-FR-LPM-13	UC-13	Level Assessment	Assessment	Assessment	Increment 2	TC-23 , TC-24

RE-FR-PM-14	UC-14	Create Project	Create project	Project	Implemented	TC-25 , TC-26 , TC-27
RE-FR-PM-15	UC-15	Edit Project	Edit project	Project	Implemented	TC-28 , TC-29
RE-FR-PM-16	UC-16	Delete Project	Delete project	Project	Implemented	TC-30 , TC-31
RE-FR-PM-17	UC-17	View Projects	View projects	Project	Implemented	TC-32
RE-FR-PM-18	UC-18	View Projects in Path	View projects	Project	Implemented	TC-32
RE-FR-PM-19	UC-19	Project Details	Details	Project	Implemented	TC-33
RE-FR-PM-20	UC-20	Start Project	Start project	Project	Implemented	TC-34 , TC-35

RE-FR-PM-21	UC-21	Project Status	Status	Project	Increment 2	—
RE-FR-PM-22	UC-22	Multi Stage	Stages	Project	Increment 2	—
RE-FR-PM-23	UC-23	Dependencies	Dependencies	Project	Increment 2	—
RE-FR-PM-24	UC-24	Filter by Level	Filtering	Project	Increment 2	—
RE-FR-PM-25	UC-25	Starter Files	Files	Project	Increment 2	—
RE-FR-PM-26	UC-26	Prerequisites	Prerequisites	Project	Increment 2	—
RE-FR-PM-27	UC-27	Code Editor	Editor	Code Editor	Increment 2	TC-36
RE-FR-PM-28	UC-28	Run Code	Execution	Execution	Increment 2	TC-37 , TC-38
RE-FR-PM-29	UC-29	Auto Save	Saving	Project	Increment 2	TC-39
RE-FR-PM-30	UC-30	Progress Bar	Progress	Project	Increment 2	—
RE-FR-PM-31	UC-31	Statistics	Stats	Statistics	Increment 2	—
RE-FR-PM-32	UC-32	Version History	Versions	Versioning	Increment 2	—
RE-FR-PM-33	UC-33	Submit Project	Submission	Submission	Increment 2	TC-43
RE-FR-PM-34	UC-34	Delete Tasks	Tasks	Tasks	Increment 2	—
RE-FR-PM-35	UC-35	Add Tasks	Tasks	Tasks	Increment 2	—
RE-FR-PM-36	UC-36	Hints	Hints	Hints	Increment 2	TC-40 , TC-41 , TC-42
RE-FR- PM-37	UC-37	Task Sidebar	UI	UI	Increment 2	—
RE-FR-PM-38	UC-38	Manual Evaluation	Evaluation	Evaluation	Increment 2	TC-44
RE-FR-PM-39	UC-39	View Submitted	Evaluation	Evaluation	Increment 2	TC-45

RE-FR-PM-40	UC-40	Filter Evaluation	Evaluation	Evaluation	Increment 2	TC-45
RE-FR-LD-41	UC-41	Dashboard	Dashboard	Dashboard	Implemented	TC-47

4.8 Initial test cases :

This section presents a set of representative test cases used to validate the main functional requirements of the Project-Based Programming Education System. The test cases cover core functions such as account management, learning paths, project management, dashboard, and advanced project-based learning features.

Sce-01: Login functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-01	Valid login	1. Open the login page. 2. Enter valid email/username and password. 3. Click Login.	The user is authenticated and redirected to the home/dashboard page.
TC-02	Login with empty fields	1. Open the login page. 2. Leave one or more required fields empty. 3. Click Login.	Validation messages are displayed beside the empty fields.
TC-03	Login with invalid credentials	1. Open the login page. 2. Enter invalid email/username or password. 3. Click Login.	An error message is displayed indicating invalid credentials.

Sce-02: Create account functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-04	Create account successfully	1. Open the registration page. 2. Enter valid required data. 3. Click Create Account.	A new account is created successfully and the user is redirected to the system.
TC-05	Create account with existing email	1. Open the registration page. 2. Enter an email already registered. 3. Fill remaining fields and submit.	An error message appears indicating that the email already exists.
TC-06	Create account with empty fields	1. Open the registration page. 2. Leave one or more required fields empty. 3. Click Create Account.	Validation messages are displayed beside the empty fields.
TC-07	Create account with invalid data	1. Open the registration page. 2. Enter invalid data in one or more fields. 3. Submit the form.	Validation messages are displayed beside the invalid fields.

Sce-03: Profile Management functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-08	View profile successfully	1. Log in with a valid account. 2. Open the profile page.	The system displays the user profile information correctly.

TC-09	Edit profile successfully	1. Open the profile page. 2. Click Edit. 3. Modify fields with valid data. 4. Click Save.	The system saves the updates and displays a success message.
TC-10	Edit profile with empty fields	1. Open edit profile page. 2. Clear a required field. 3. Click Save.	The system prevents saving and displays a validation message.
TC-11	Edit profile with invalid data	1. Open edit profile page. 2. Enter invalid data. 3. Click Save.	The system displays an error message beside the invalid field.

Sce-04: Add Learning Path functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-12	Add learning path successfully	1. Log in as instructor. 2. Open Add Learning Path page. 3. Enter valid path data. 4. Click Save.	The learning path is created and appears in the learning paths list.
TC-13	Add learning path with empty fields	1. Open Add Learning Path page. 2. Leave required fields empty. 3. Click Save.	Validation messages are displayed beside empty fields.
TC-14	Add duplicate learning path	1. Open Add Learning Path page. 2. Enter an existing path title. 3. Click Save.	The system displays a message indicating that the path already exists.

Sce-05: Edit Learning Path functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-15	Edit learning path successfully	1. Log in as instructor. 2. Select an existing learning path. 3. Click Edit. 4. Modify valid data and save.	The learning path is updated successfully.
TC-16	Edit learning path with invalid data	1. Open edit path page. 2. Enter invalid data. 3. Click Save.	The system displays validation messages and prevents saving.

Sce-06: Delete Learning Path functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-17	Delete learning path successfully	1. Log in as instructor. 2. Open learning paths list. 3. Click Delete for a path. 4. Confirm deletion.	The learning path is deleted and removed from the list.
TC-18	Cancel learning path deletion	1. Click Delete for a learning path. 2. Cancel the confirmation message.	The learning path remains unchanged.

Sce-07: View Learning Path functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
--------------	-----------------	------------	-----------------

TC-19	View available learning paths	1. Log in as learner or instructor. 2. Open Learning Paths page.	The system displays the list of available learning paths.
TC-20	View learning path details	1. Open Learning Paths page.	The system displays path details such as title, description,
		2. Select a specific learning path.	level, and related projects.

Sce-08: Join Learning Path functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-21	Join learning path successfully	1. Log in as learner. 2. Open a learning path. 3. Click Join.	The learner is added to the selected path and the path appears in the learner dashboard.
TC-22	Join already joined path	1. Log in as learner. 2. Open a path already joined. 3. Click Join again.	The system prevents duplicate joining or displays an appropriate message.

Sce-09: Learner Level Assessment functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-23	Complete level assessment	1. Join a learning path. 2. Answer assessment questions. 3. Submit answers.	The system calculates the result and assigns a suitable learner level.
TC-24	Cancel level assessment	1. Open level assessment page. 2. Close or skip the assessment.	The system informs the learner that level assessment is required to access suitable projects.

Sce-10: Add Project functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-25	Add project successfully	1. Log in as instructor. 2. Open a learning path. 3. Click Add Project. 4. Enter valid project details. 5. Save.	The project is created and appears inside the selected learning path.
TC-26	Add project with empty fields	1. Open Add Project page. 2. Leave required fields empty. 3. Click Save.	Validation messages are displayed beside empty fields.
TC-27	Add duplicate project	1. Open Add Project page. 2. Enter an existing project title. 3. Click Save.	The system displays a message indicating that the project already exists.

Sce-11: Edit Project functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-28	Edit project successfully	1. Log in as instructor. 2. Open project list. 3. Click Edit for a project. 4. Modify data and save.	The project information is updated successfully.
TC-29	Edit project with invalid data	1. Open Edit Project page. 2. Enter invalid data. 3. Click Save.	The system displays validation messages and prevents saving.

Sce-12: Delete Project functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-30	Delete project successfully	1. Log in as instructor. 2. Open project list. 3. Click Delete. 4. Confirm deletion.	The project is deleted and removed from the project list.
TC-31	Cancel project deletion	1. Click Delete for a project. 2. Cancel the	The project remains unchanged.
		confirmation message.	

Sce-13: View Projects functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-32	View projects in learning path	1. Open a learning path. 2. Go to projects section.	The system displays all projects related to the selected learning path.
TC-33	View project details	1. Select a project from the list.	The system displays project title, description, tasks, requirements, resources, and level.

Sce-14: Start Project functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
--------------	-----------------	------------	-----------------

TC-34	Start project successfully	1. Log in as learner. 2. Open a joined learning path. 3. Select a project. 4. Click Start.	The project status becomes In Progress and the project workspace opens.
TC-35	Start locked project	1. Open a project with incomplete prerequisites. 2. Click Start.	The system prevents access and displays the required prerequisite project.

Sce-15: Code Editor and Execution functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-36	Open code editor	1. Start a project. 2. Open the project workspace.	The system displays the code editor and starter files.
TC-37	Run code successfully	1. Write valid code in the editor. 2. Click Run.	The system executes the code and displays the output.
TC-38	Run code with error	1. Write code containing an error. 2. Click Run.	The system displays the error message in the output panel.
TC-39	Auto save progress	1. Modify code in the editor. 2. Wait for auto-save period or leave the page.	The system saves the latest project state automatically.

Sce-16: Hints functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
--------------	-----------------	------------	-----------------

TC-40	Request first hint	1. Open project workspace. 2. Click Request Hint.	The system displays the first hint without revealing the full solution.
TC-41	Request additional hints	1. Click Request Hint multiple times.	The system displays hints gradually from general to more specific.
TC-42	No more hints available	1. Request hints after all hints are displayed.	The system informs the learner that no more hints are available.

Sce-17: Project Submission and Evaluation functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-43	Submit project successfully	1. Complete project tasks. 2. Click Submit.	The system submits the project and changes its status to Submitted.
TC-44	Manual evaluation by instructor	1. Log in as instructor. 2. Open submitted projects. 3. Add feedback and rating. 4. Save evaluation.	The evaluation is saved and visible to the learner.
TC-45	Filter submitted projects	1. Open submitted projects list. 2. Apply evaluation status filter.	The system displays projects according to their evaluation status.

Sce-18: Dashboard functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
--------------	-----------------	------------	-----------------

TC-46	View learner dashboard	1. Log in as learner. 2. Open Dashboard.	The system displays enrolled paths, active projects, completed projects, and progress.
TC-47	Update dashboard after progress	1. Start or complete a project. 2. Return to Dashboard.	The dashboard statistics are updated according to the latest progress.

4.8 Summary :

This chapter presented the system requirements analysis, including functional and non-functional requirements. It also introduced the use cases, UML diagrams, and the selected development methodology modeling. In addition, test cases and the Requirements Traceability Matrix (RTM) were provided to ensure requirement coverage and system validation.

Chapter 5 : Architectural design of the system

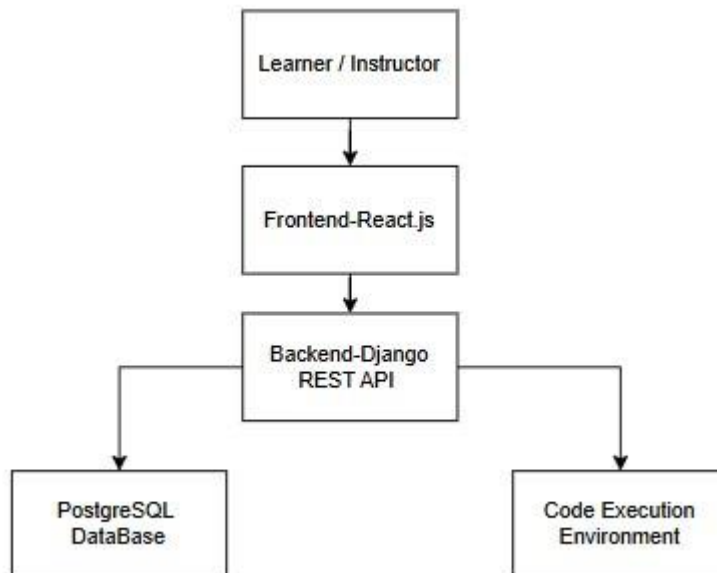
5.1 High-Level Design :

The architectural block diagram shows the main components of the system and how they interact with each other. The system follows a client-server architecture, where the frontend communicates with the backend through APIs, and the backend interacts with the database.

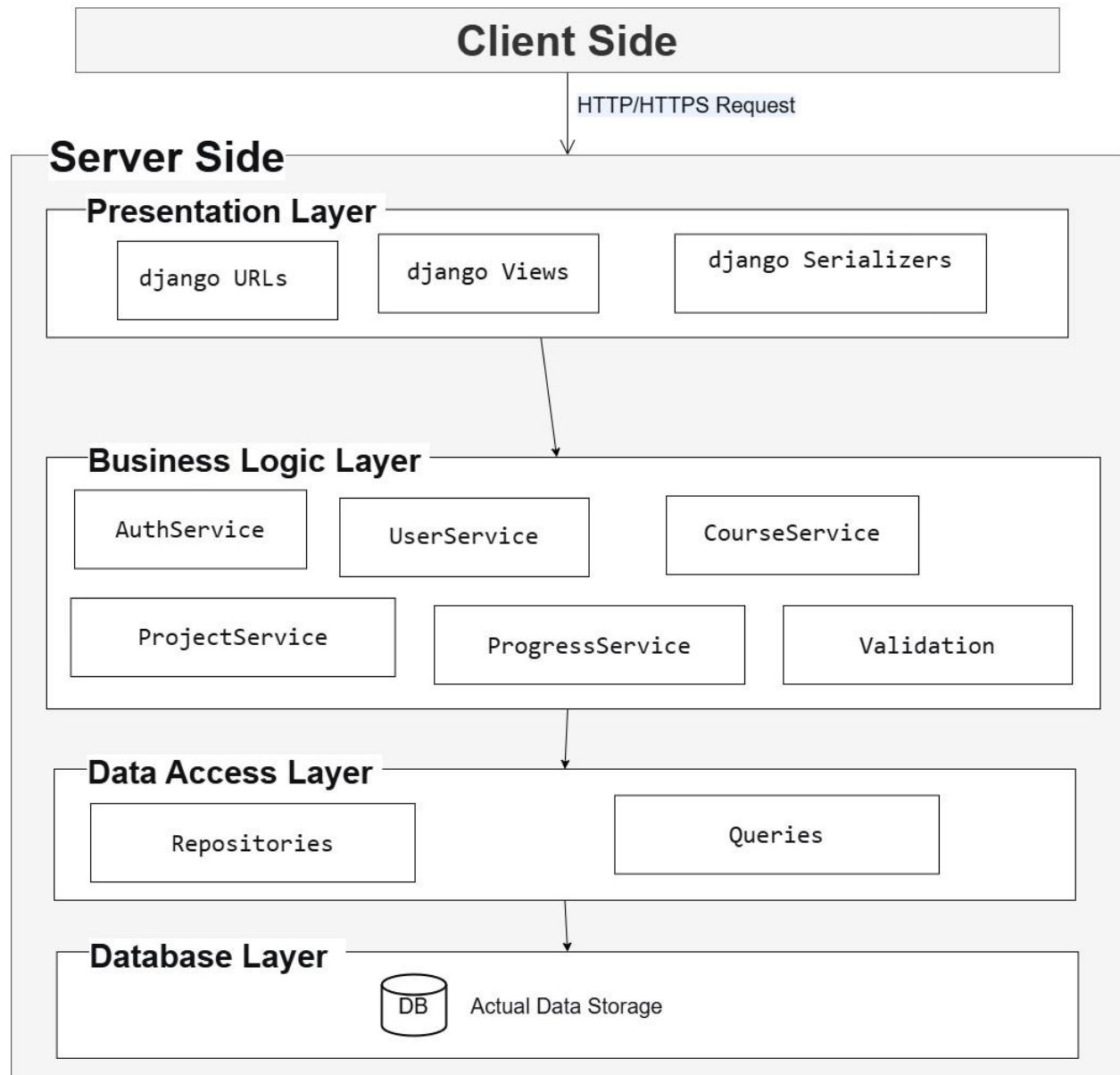
The main components are:

- User Interface (Frontend): Built with React and used by learners and instructors.
- Backend Server: Built with Django and responsible for business logic and API handling.
- Database: PostgreSQL is used to store users, learning paths, projects, submissions, and progress.
- Code Execution Environment: Responsible for running learner code and returning the output.
- Authentication Module: Handles login, registration, and user permissions.

5.1.1 System Block Diagram :



5.1.1 Architectural design style used :



5.1.1 System Architecture Design :

The system is designed using a layered architecture that separates the application into multiple layers, each with a specific responsibility.

The **presentation layer** handles incoming requests through Django URLs, views, and serializers, and communicates with the frontend using APIs.

The **business logic layer** contains the core functionalities of the system, such as authentication, user management, learning paths, project management, and validation.

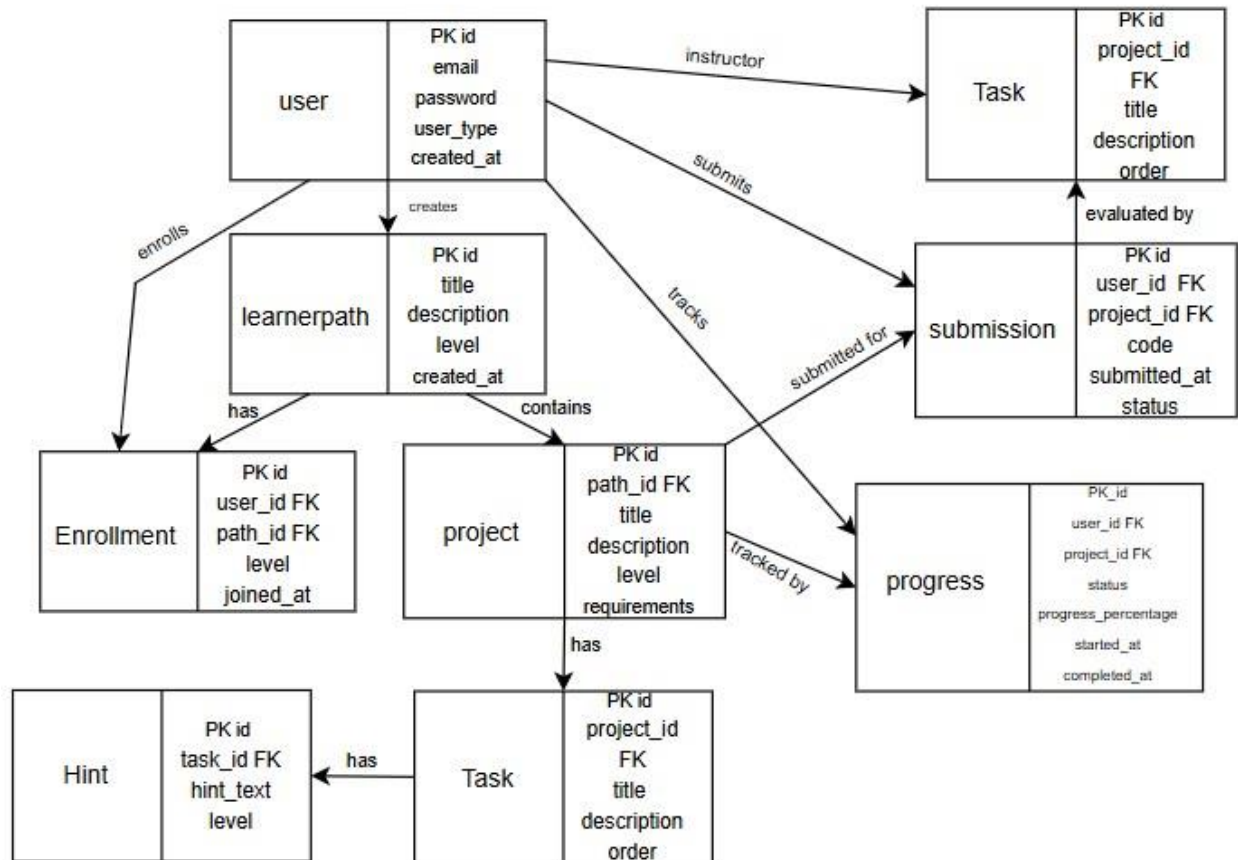
The data **access layer** is responsible for interacting with the database through queries and data handling operations.

Finally, the **database layer** stores all system data, including users, projects, learning paths, and progress.

This design improves system organization, maintainability, and scalability by clearly separating concerns between different components.

5.1 DataBase Design :

5.1.1 ER Diagram :



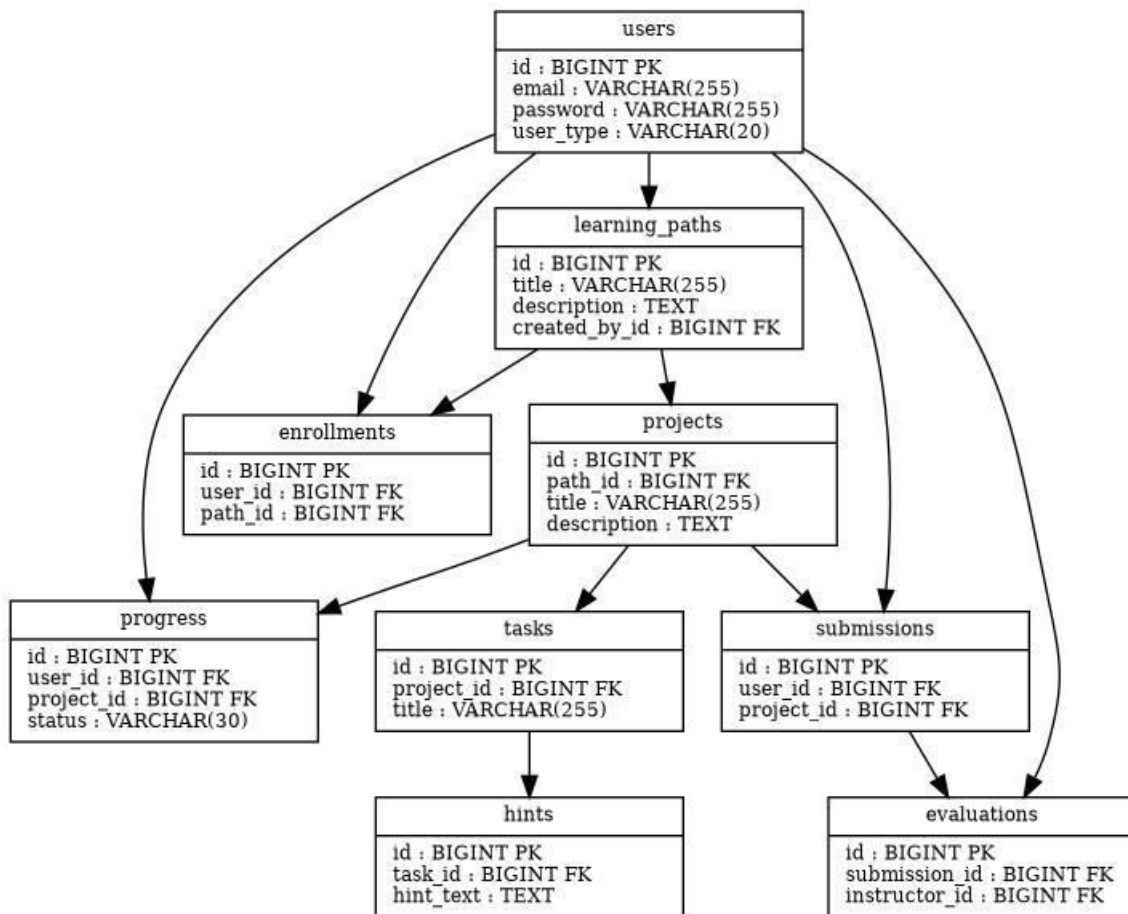
The ERD represents the main entities of the system and their relationships .

It shows how users interact with learning paths , projects , tasks , progress ,

Submissions , evaluation , and hints .

This model helps organize the database structure and supports the main system functionalities .

5.1.1 Relational Schema :



The relational schema represents the database tables, fields, data types, primary keys, and foreign keys. It shows how the main entities are stored and connected in the database to support users, learning paths, projects, progress, submissions, evaluations, and hints.

5.1.1 Constraints and Keys :

Database constraints are used to ensure data integrity and consistency within the system. The database uses several types of keys and constraints such as primary keys, foreign keys, unique constraints, and validation checks.

Constraint Type	Example	Description
Primary Key	id	Used to uniquely identify each record in the table.
Foreign Key	user_id, project_id, path_id	Used to create relationships between tables.
Unique Constraint	email	Prevents duplicate user email addresses.
Check Constraint	progress_percentage >= 0	Ensures valid values are stored in the database.

5.1 External APIs Design :

The system uses RESTful APIs to enable communication between the frontend and backend. APIs are responsible for handling authentication, learning paths, project management, progress tracking, and code execution.

Each API endpoint uses HTTP methods such as GET and POST to send and receive data in JSON format.

API Endpoint	Method	Description	Response
/login/	POST	Authenticate user login	JWT Token / Success Message
/register/learner/	POST	Create learner account	User Data
/profile/	GET	Retrieve user profile	User Profile Data
/learner/dashboard/	GET	Display learner dashboard	Dashboard Data
/quiz/submit/	POST	Submit quiz answers	Evaluation Result
/projects/	GET	Retrieve projects list	Projects Data
/projects/{id}/work/	POST	Submit code execution	Output / Errors

5.1 Summary :

This chapter explained the architectural and database design of the system. It presented the system architecture, layered design approach, ER diagram, relational schema, constraints, and API design. These designs provide a clear structure that supports system scalability, maintainability, and efficient data management.

Chapter 6 : Execution environment and tools

6.1 Tools Used :

- **Backend:** Django
- **Frontend:** React.js
- **Database:** PostgreSQL
- **Version Control:** Git & GitHub
- **Code Editor:** Visual Studio Code
- **API Testing:** Postman
- **Browser:** Google Chrome

6.2 Development Environment and Project Structure :

The system was developed in a web-based environment using Django for the backend and React for the frontend.

The project follows a modular structure divided into:

- Backend (business logic and database)
- Frontend (user interface)

6.3 Programming Languages and Frameworks :

- Programming Languages: Python, JavaScript, HTML, CSS
- Frameworks: Django (Backend), React.js (Frontend)

6.4 Database Server and Operating System :

- Database Server: PostgreSQL
- Operating System: Windows

6.5 Dependencies :

- Backend: Django, Django REST Framework
- Frontend: React.js, Axios
- Others: JSON, REST APIs

6.6 Version Control and Source Code Management :

- Version Control System: Git
- Platform: GitHub

6.7 Continuous Integration and Deployment (CI/CD):

CI/CD tools were not applied in this project. The development, testing, and deployment processes were performed manually.

6.8 Testing Environment :

The system was tested in a local development environment using a web browser. Manual testing was performed to verify system functionality and user interactions.

6.9 Summary :

This chapter introduced the development environment, tools, programming languages, frameworks, database server, dependencies, and testing environment used in the project. The selected technologies provided a suitable environment for implementing and testing the proposed system efficiently.

Chapter 7 : Programmatic implementation

7.1 Project Structure :

JUNIORPROJECT-MAIN/

```
|
|
|—— buildly---project-Based-learning-platform/
|  |
|  |—— backendPBL/
|  |  |—— projectBPL/
|  |  |—— check_backend.bat
|  |  |—— recreate_venv.bat / .ps1
|  |  |—— run_backend.ps1
|  |  |—— start_backend.bat / .ps1
|  |  └── test_backend.md
|  |
|  |—— docker/
|  |  └── python-runner/
|  |    └── Dockerfile
|  |
|  |—— frontend/
|  |  |—— src/
|  |  |—— index.html
|  |  |—— package.json
```

```
| | |— package-lock.json
| | |— vite.config.js
| | |— run_frontend.ps1
| |
| |— README.md
| |— junior_project.zip
| |— PBL.pdf
```

The project is organized into a modular structure consisting of backend and frontend components.

The backend (backendPBL) handles the server-side logic and APIs, while the frontend manages the user interface.

Additional folders such as Docker are used for execution environments, and configuration and documentation files are included for project management.

7.2 Modules Implementation :

The system was implemented using several main modules, where each module is responsible for a specific functionality.

1. Authentication Module

This module handles user registration, login, logout, and user authentication. It ensures that only authorized users can access the system features.

2. Profile Management Module

This module allows users to view and update their personal information.

3. Learning Path Module

This module enables instructors to create, edit, delete, and view learning paths. It also allows learners to browse and join available learning paths.

4. Project Management Module

This module allows instructors to create and manage projects inside learning paths. Learners can view project details, start projects, and track their progress.

5. Code Editor Module

This module provides an integrated code editor that allows learners to write code, run it, and view the output inside the platform.

6. Dashboard Module

This module displays learner-related information such as enrolled learning paths, started projects, progress, and learning statistics.

7. Evaluation Module

This module allows submitted projects to be reviewed and evaluated, either automatically or manually by the instructor

7.3 API Implementation :

The system uses REST APIs to enable communication between the frontend and backend. These APIs handle user authentication and data exchange.

Example : Register learner API

<u>Item</u>	<u>Description</u>
<u>API Name</u>	<u>Register Learner</u>
<u>Method</u>	<u>POST</u>
<u>URL</u>	<u>/register/learner</u>
<u>Input</u>	<u>Email , password , password2</u>
<u>Output</u>	<u>User data and authentication tokens</u>

```

class RegisterLearnerView(generics.CreateAPIView):
    """إنشاء حساب متعلم"""
    queryset = CustomUser.objects.all()
    serializer_class = RegisterLearnerSerializer
    permission_classes = [permissions.AllowAny]

    def create(self, request, *args, **kwargs):
        serializer = self.get_serializer(data=request.data)
        serializer.is_valid(raise_exception=True)
        user = serializer.save()

        # إنشاء توكنات JWT
        refresh = RefreshToken.for_user(user)

        # تحضير بيانات المستخدم حسب النوع
        user_data = {
            'id': user.id,
            'email': user.email,
            'user_type': user.get_user_type_display(),
        }

        # إضافة بيانات المسارات فقط للمتعلمين
        if user.is_learner:
            user_data['enrolled_courses_count'] = user.get_enrolled_courses_count()
            user_data['enrolled_courses_titles'] = user.get_enrolled_courses_list()

        return Response({
            'message': _('تم إنشاء حساب المتعلم بنجاح'),
            'user': user_data,
            'tokens': {
                'refresh': str(refresh),
                'access': str(refresh.access_token),
            }
        }, status=status.HTTP_201_CREATED)

```

This API is responsible for registering a new learner. It receives user data from the frontend, validates the input using a serializer, creates a new user, and returns authentication tokens using JWT.

7.3 Data Access Layer Implementation:

The data access layer is implemented using Django ORM. This layer is responsible for interacting with the database without writing raw SQL queries.

The system uses Django models to represent database tables, and ORM queries are used to create, retrieve, and manage data.

Example ORM Query :

```
queryset = CustomUser.objects.all()
```

This query retrieves all users from the CustomUser table .

Another example is checking whether an email already exists :

```
CustomUser.objects.filter(email=attrs['email']).exists():
```

This query checks if the entered email is already registered in the database .

Using Django ORM makes database operations easier, safer, and more maintainable. It also helps reduce direct dependency on SQL queries and improves code readability.

7.5 Business Logic Layer Implementation

The business logic layer contains the rules and conditions that control how the system behaves. In this project, the business logic is implemented mainly inside views and serializers.

Example : Email Validation login :

```

if CustomUser.objects.filter(email=attrs['email']).exists():
    raise serializers.ValidationError(
        {"email": ("البريد الإلكتروني موجود مسبقاً")}
    )

```

This logic checks whether the email is already registered before creating a new account .

Example : Password Matching Logic :

```

if attrs['password'] != attrs['password2']:
    raise serializers.ValidationError(
        {"password": ("كلمات المرور غير متطابقة")}
    )

```

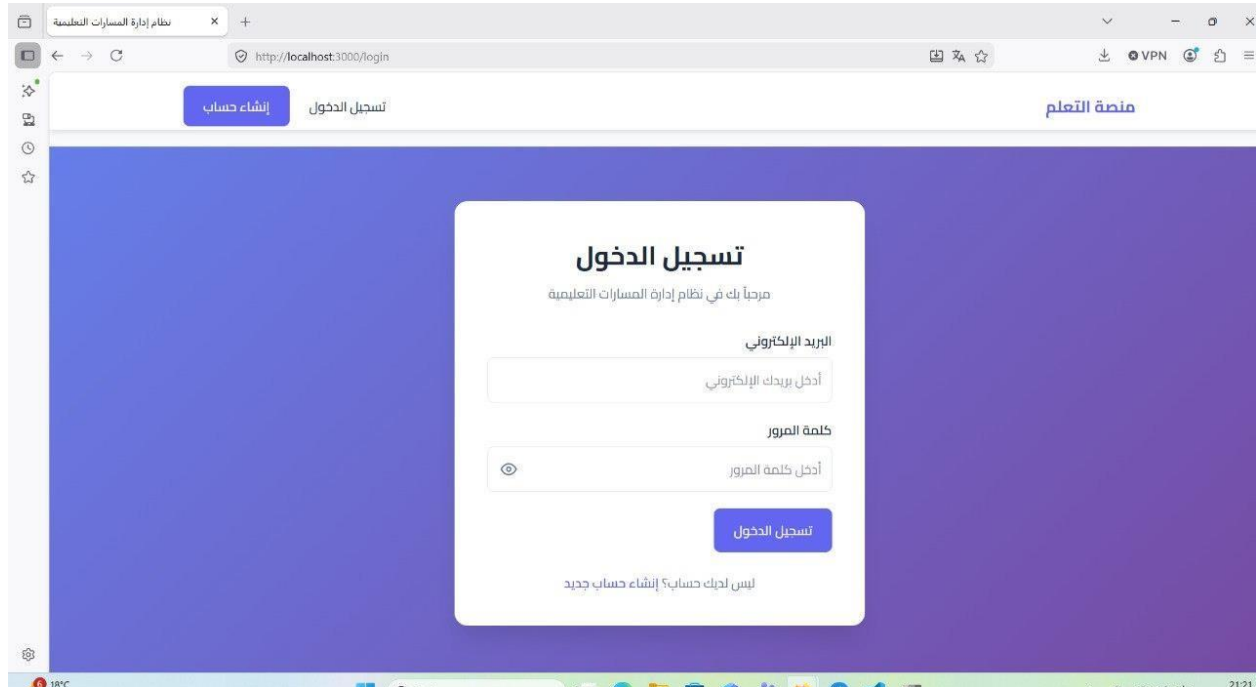
This logic ensures that the password and confirmation password are identical before completing registration .

These rules help prevent invalid data from being stored in the database and ensure that user registration follows the required system conditions.

7.6 User Interface Implementation :

The user interface was implemented using React.js. It provides interactive pages that allow users to navigate through the system easily and perform the main functions.

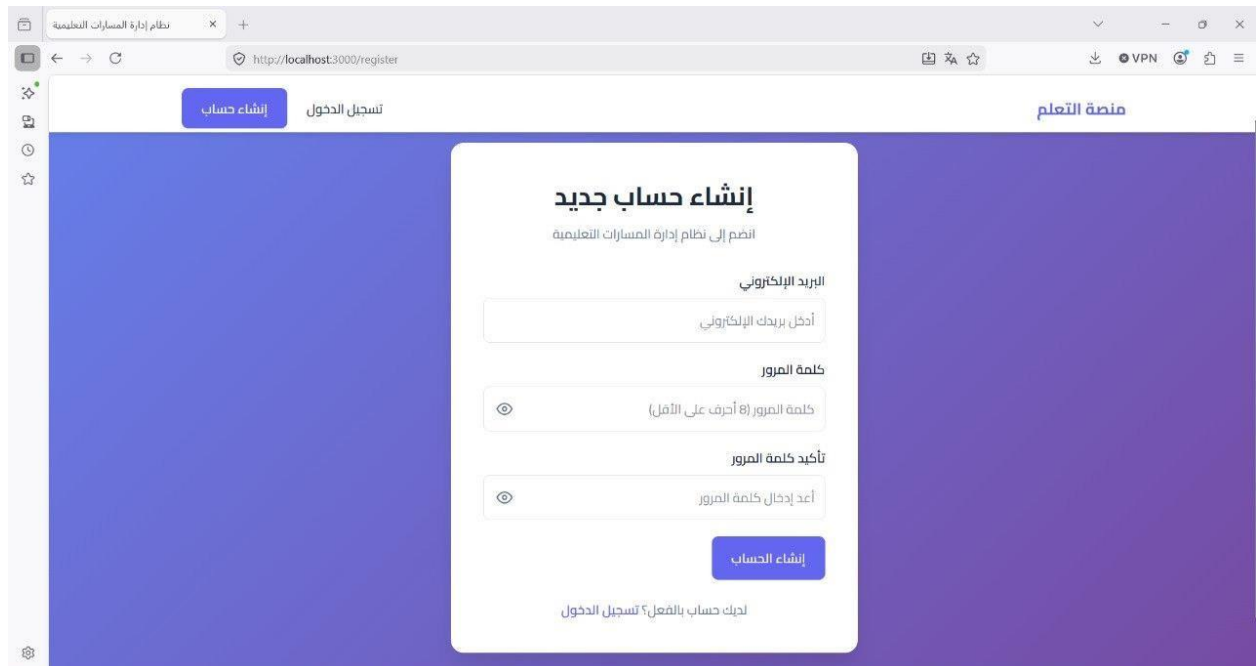
Login Page :



The screenshot shows a web browser window with the URL `http://localhost:3000/login`. The page has a dark blue header with the text "منصة التعلم" (Learning Platform) on the right. Below the header, there are two buttons: "إنشاء حساب" (Create Account) and "تسجيل الدخول" (Login). The main content area has a light blue background with a white box in the center. The box contains the title "تسجيل الدخول" (Login) and the subtitle "مرحباً بك في نظام إدارة المسارات التعليمية" (Welcome to the Educational Path Management System). There are two input fields: "البريد الإلكتروني" (Email) and "كلمة المرور" (Password). Below the password field is a blue button labeled "تسجيل الدخول" (Login). At the bottom of the box, there is a link: "ليس لديك حساب؟ إنشاء حساب جديد" (Don't have an account? Create a new account).

This page allows users to log in using their email and password.

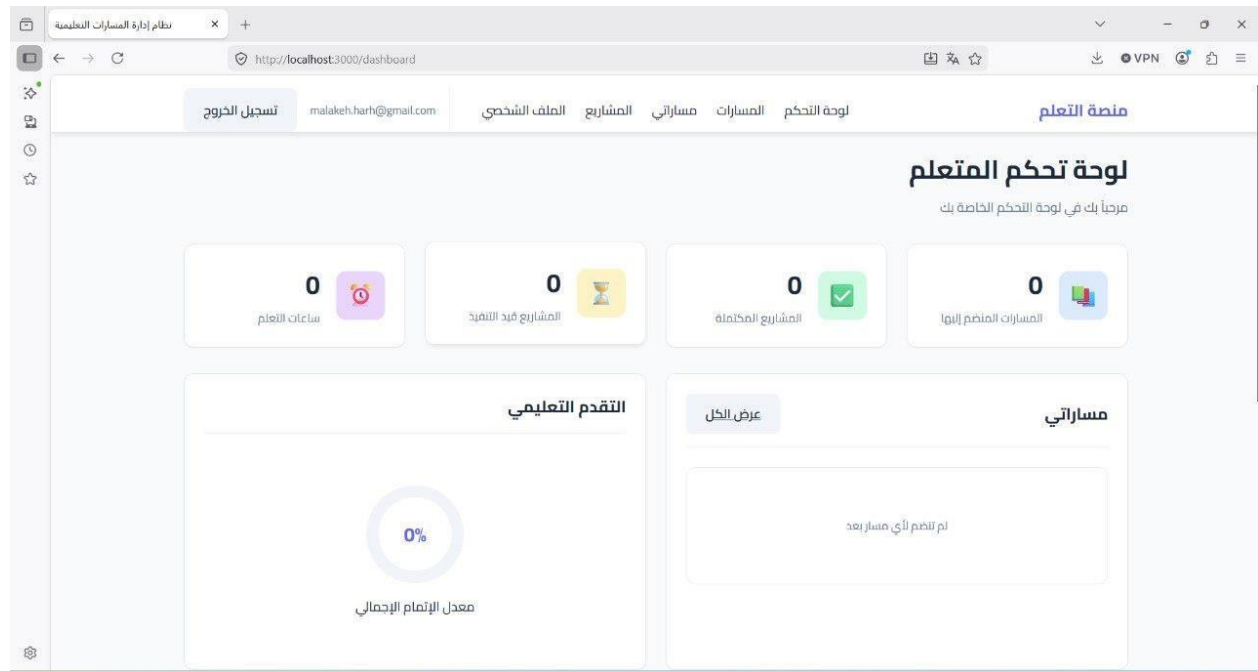
Register Page:



The screenshot shows a web browser window with the URL `http://localhost:3000/register`. The page has a dark blue header with the text "منصة التعلم" (Learning Platform) on the right. Below the header, there are two buttons: "إنشاء حساب" (Create Account) and "تسجيل الدخول" (Login). The main content area has a light blue background with a white box in the center. The box contains the title "إنشاء حساب جديد" (Create New Account) and the subtitle "انضم إلى نظام إدارة المسارات التعليمية" (Join the Educational Path Management System). There are three input fields: "البريد الإلكتروني" (Email), "كلمة المرور" (Password), and "تأكيد كلمة المرور" (Confirm Password). Below the password field is a blue button labeled "إنشاء الحساب" (Create Account). At the bottom of the box, there is a link: "لديك حساب بالفعل؟ تسجيل الدخول" (Already have an account? Login).

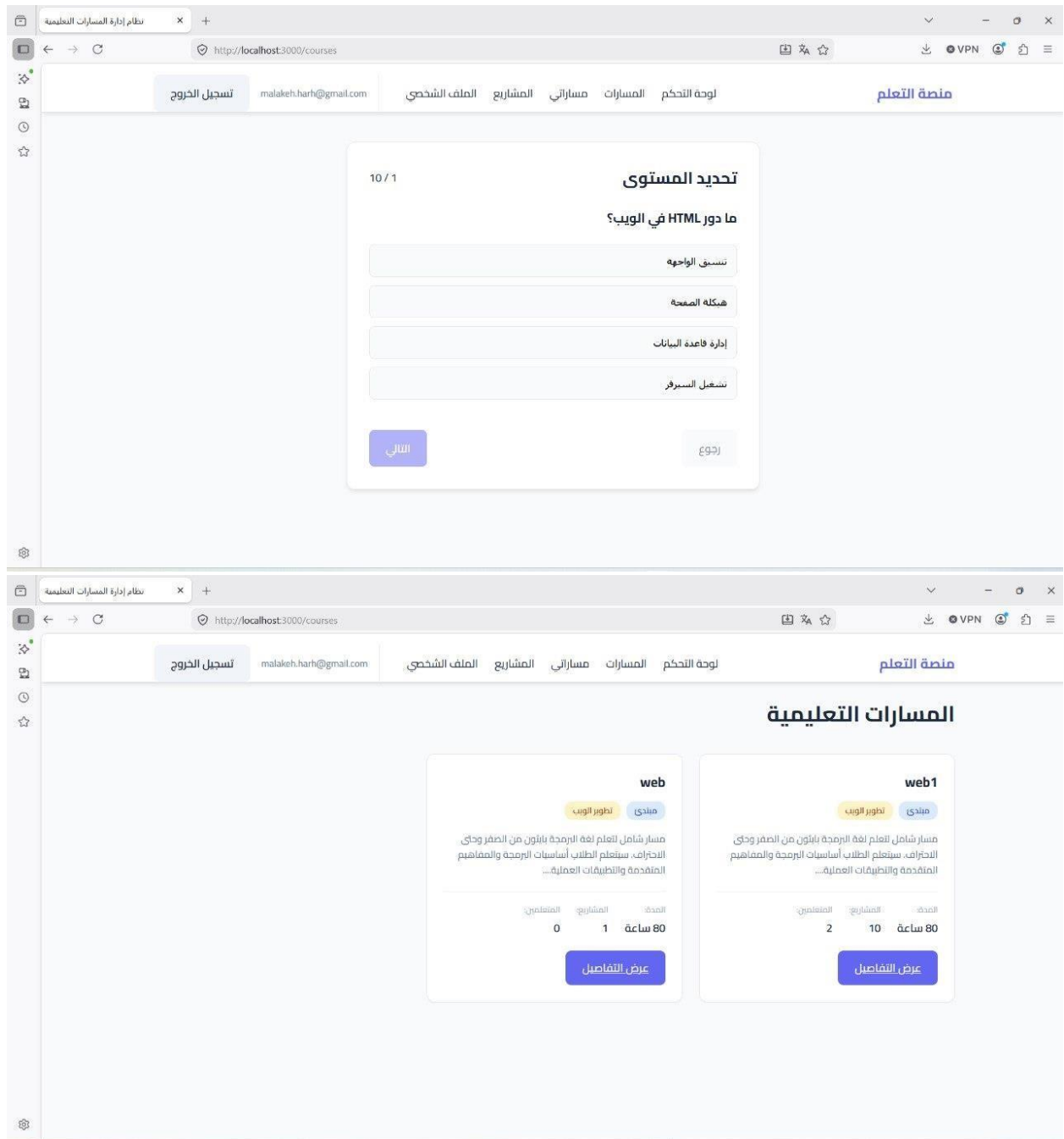
This page allows new users to create an account as a learner or instructor.

Learner Dashboard :



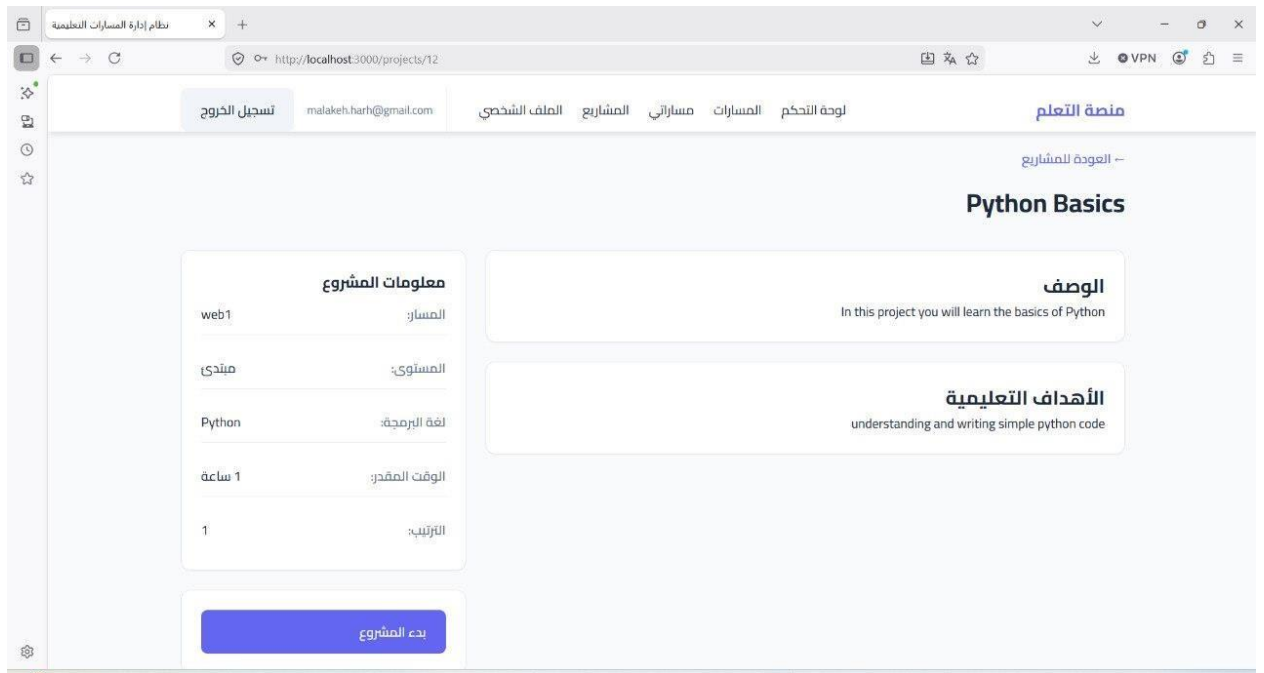
This page displays the learner's enrolled paths, active projects, and progress.

Learning Paths Page :



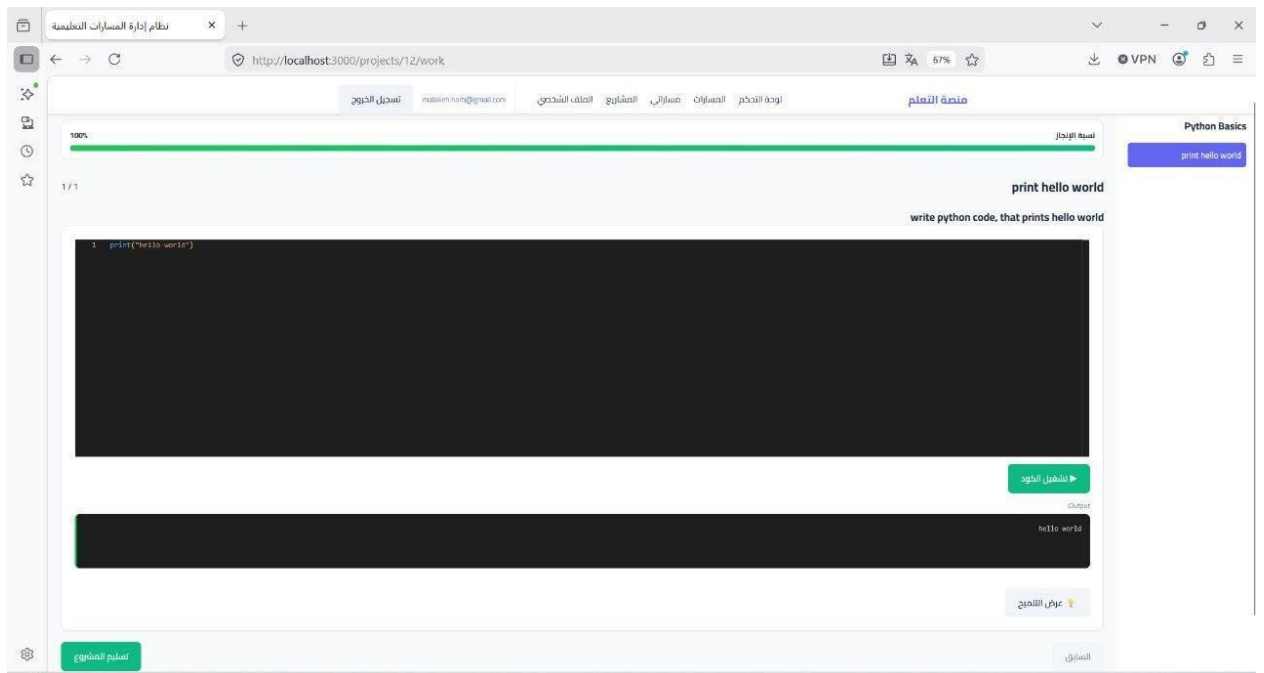
This page shows the available learning paths that learners can browse and join.

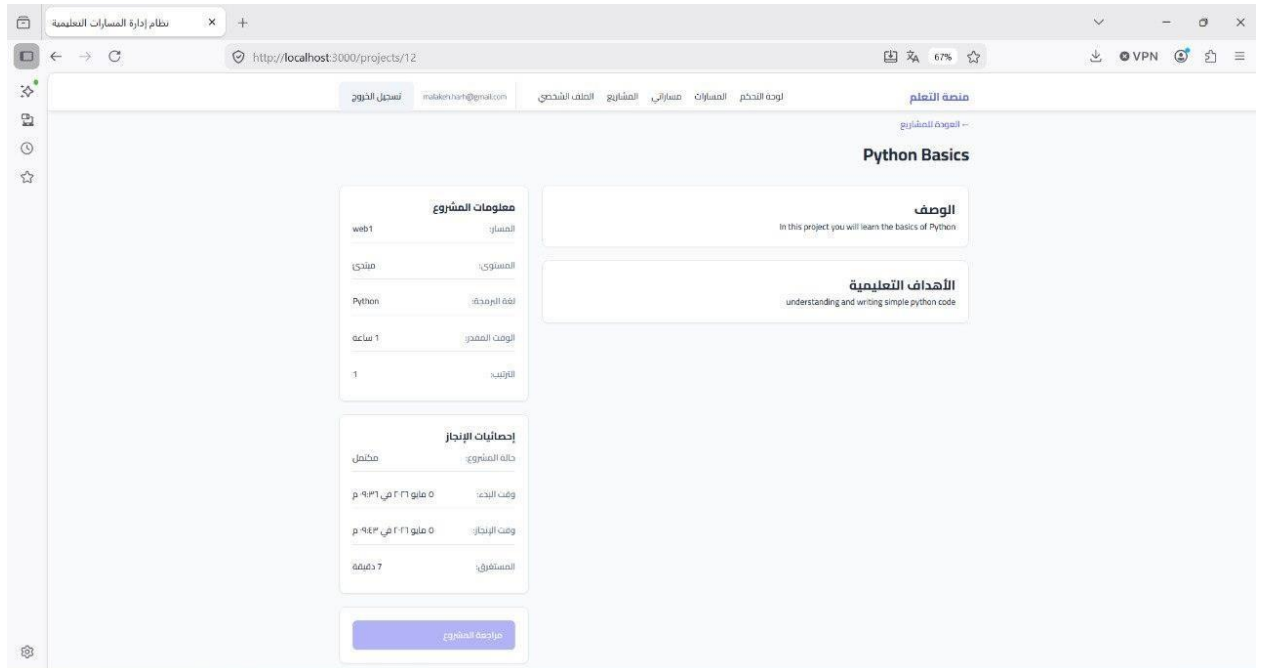
Project Details Page :



This page displays project information such as tasks, requirements, and resources.

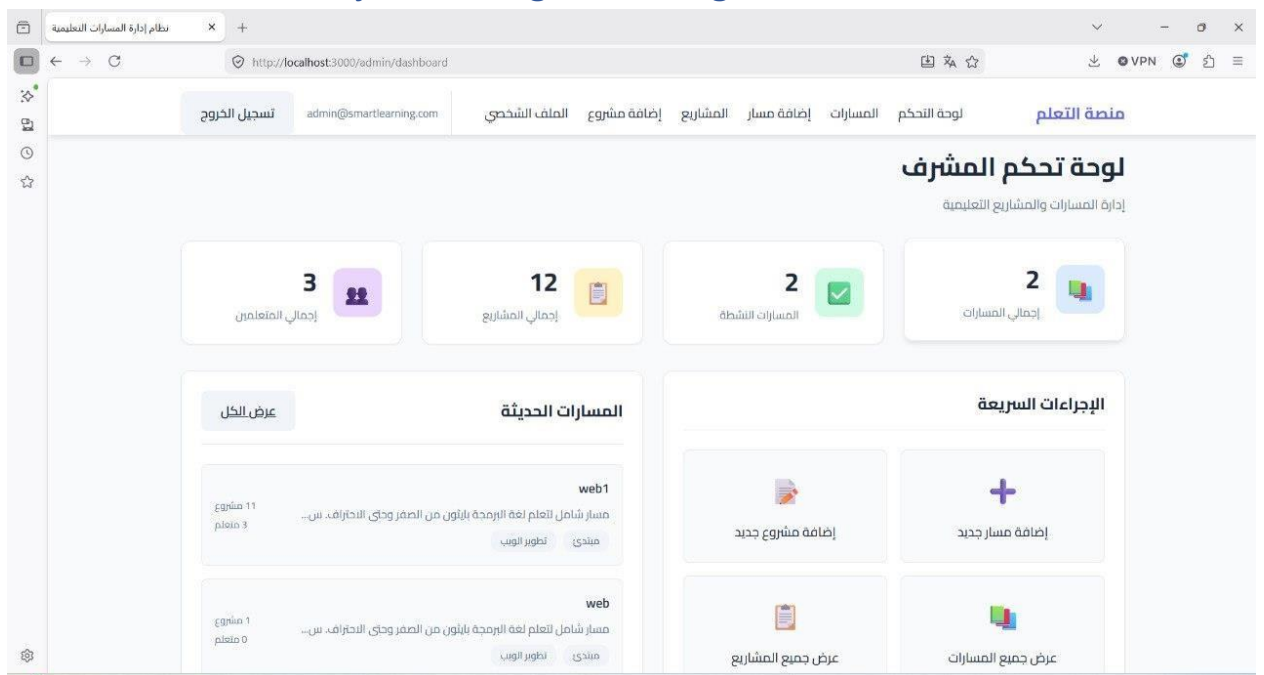
Code Editor Page :





This page allows learners to write, run, and test their code inside the platform.

Instructor Dashboard / Project Management Page :



This page allows the instructor to manage learning paths and projects.

The following screenshots show the main user interfaces implemented in the system. Each interface was designed to be simple, clear, and easy to use for both learners and instructors.

7.7 Intelligent Algorithms Implementation :

This project does not include machine learning models or intelligent algorithms such as model training, text processing, or data analysis.

The system mainly focuses on implementing a project-based learning platform using web development technologies. Therefore, no AI algorithm was implemented in this phase.

7.7 Summary :

This chapter discussed the implementation details of the system, including the project structure, modules, APIs, data access layer, business logic, and user interface. The implementation successfully integrated the frontend and backend components to provide a functional and interactive learning platform.

Chapter 8 : Testing and evaluation

8.1 Test plan :

The test plan defines the testing process used to ensure that the system functions correctly and meets the specified requirements.

What will be tested :

The system functionalities will be tested, including:

- User authentication (login, registration)
- Learning path management
- Project management
- Code editor and execution
- Dashboard and user interactions

When testing will be performed:

Testing will be conducted during the development process after implementing each module (increment). Final testing will be performed after completing the system.

Testing tools :

Manual testing is used as the main approach, supported by:

- Web browser (Google Chrome) for UI testing
- Postman for API testing

8.2 Types Of Testing :

The following types of testing were applied in this project:

Integration Testing:

Used to verify communication between frontend and backend through APIs.

System Testing:

The full system was tested to ensure all features work together correctly.

Acceptance Testing:

The system was validated based on the project requirements.

Usability Testing:

The user interface was tested to ensure ease of use.

Compatibility Testing:

The system was tested using a web browser (Google Chrome).

8.3 Test Case Diagrams :

8.3.1 Test Case Diagrams :

Test case diagrams represent the testing scenarios used to validate system functionalities. In this project, test cases are presented in tabular form, where each test case includes steps and expected results.

Sce-04: Add Learning Path functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-12	Add learning path successfully	1. Log in as instructor. 2. Open Add Learning Path page. 3. Enter valid path data. 4. Click Save.	The learning path is created and appears in the learning paths list.
TC-13	Add learning path with empty fields	1. Open Add Learning Path page. 2. Leave required fields empty. 3. Click Save.	Validation messages are displayed beside empty fields.
TC-14	Add duplicate learning path	1. Open Add Learning Path page. 2. Enter an existing	The system displays a message indicating that the path already exists.

		path title. 3. Click Save.	
--	--	-------------------------------	--

Sce-15: Code Editor and Execution functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-36	Open code editor	1. Start a project. 2. Open the project workspace.	The system displays the code editor and starter files.
TC-37	Run code successfully	1. Write valid code in the editor. 2. Click Run.	The system executes the code and displays the output.
TC-38	Run code with error	1. Write code containing an error. 2. Click Run.	The system displays the error message in the output panel.

Sce-16: Hints functionalities

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-40	Request first hint	1. Open project workspace. 2. Click Request Hint.	The system displays the first hint without revealing the full solution.
TC-41	Request additional hints	1. Click Request Hint multiple times.	The system displays hints gradually from general to more specific.
TC-42	No more hints available	1. Request hints after all hints are displayed.	The system informs the learner that no more hints are available.

Additional test cases covering other system functionalities are defined but not fully presented in this report due to space limitations.

8.3.2 Test Results :

The test results show that the system functions correctly for most test cases.

- Most test cases passed successfully
- Minor errors were identified and fixed during testing
- The system performs as expected under normal usage

The condition	Num
Passed	44
Failed	3
Fixed	3

8.4 Test Results and Analysis :

The testing process was conducted using a set of predefined test cases to validate the system functionality.

Most test cases were successfully passed, indicating that the system works correctly under normal conditions. A small number of test cases initially failed due to minor issues such as input validation errors and user interface inconsistencies. These issues were identified and fixed during the testing phase.

The condition	Num
Passed	44
Failed	3
Fixed	3

The results show that the system is stable and meets the main functional requirements. Core functionalities such as authentication, learning path management, and project handling performed as expected.

8.7 Summary :

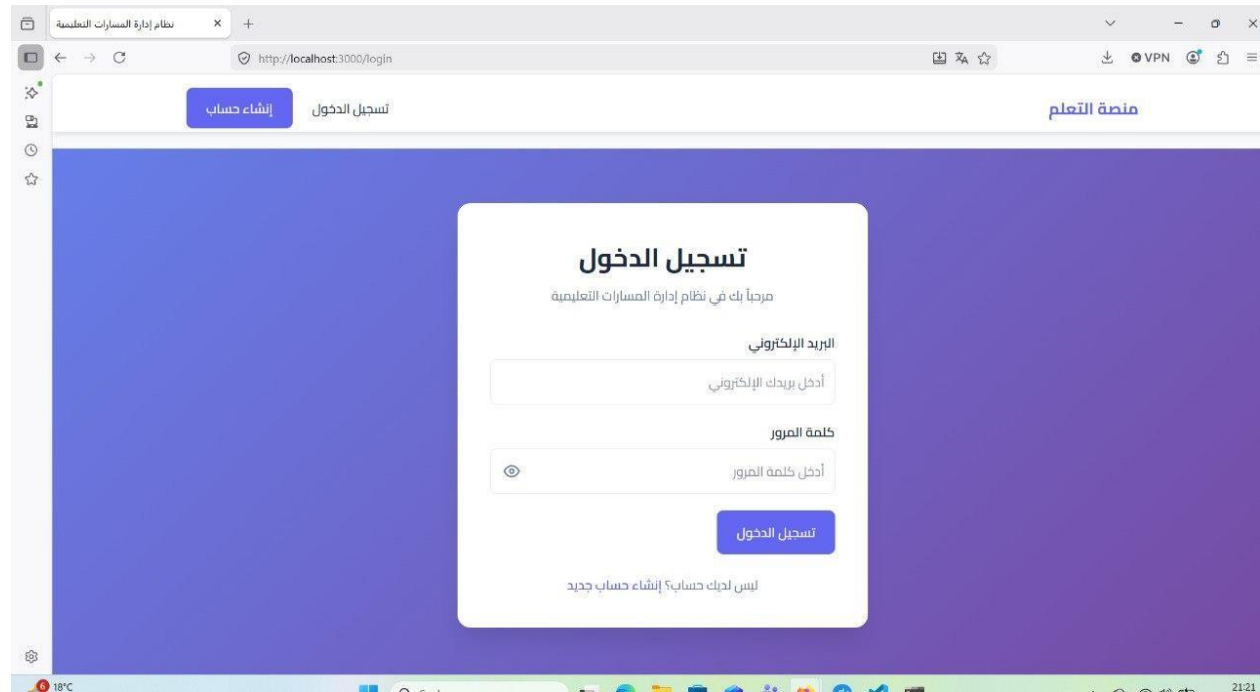
This chapter presented the testing and evaluation process of the system. Different types of testing, test cases, quality metrics, and user testing were discussed. The results showed that the system performs correctly and satisfies the main functional requirements with stable performance.

Chapter 9 : Results and analysis

9.1 System implementation results :

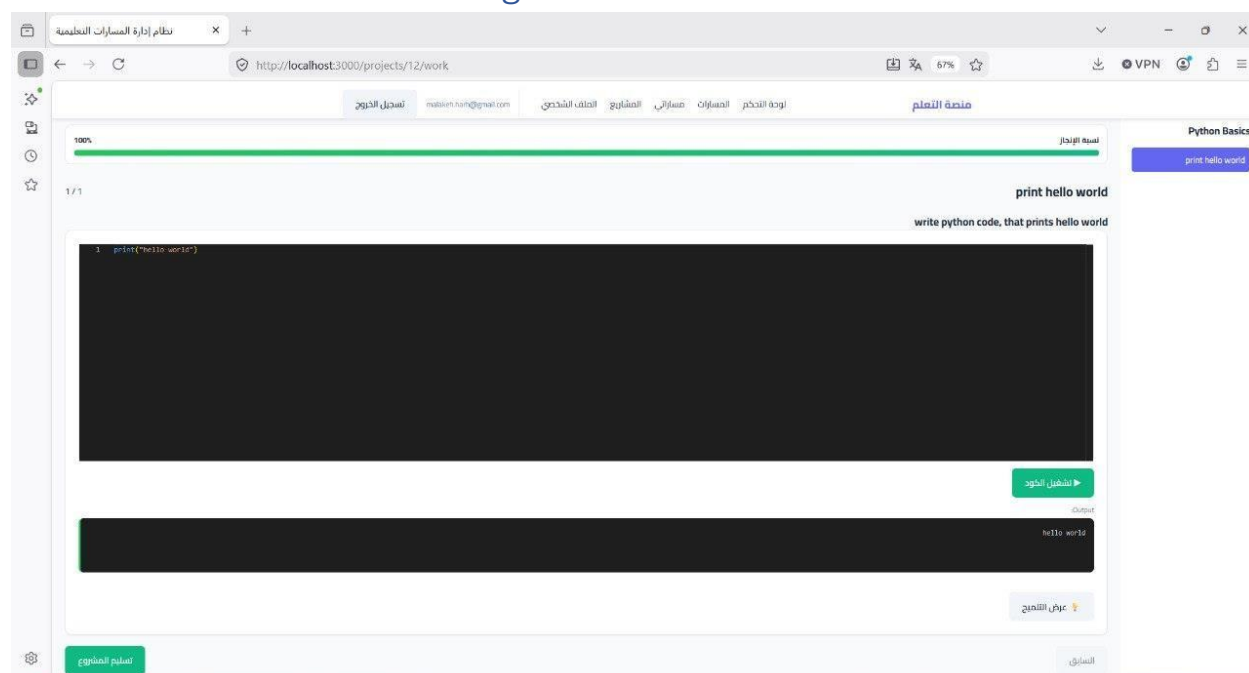
The following screenshots illustrate the main functionalities of the system. They demonstrate how users interact with different features such as authentication and project execution.

Login page :



This interface allows users to log in by entering their email and password. It ensures secure access to the system and provides a simple and user-friendly authentication process.

Code Editor and Execution Page :



This interface provides an integrated code editor where learners can write, run, and test their code. The system displays the output directly, allowing learners to practice programming in an interactive environment.

9.2 Comparative Analysis with Existing Systems:

This section presents a comparison between the proposed system and existing programming learning platforms. The comparison focuses on key aspects such as speed, accuracy, and cost.

Feature	Proposed System	Existing Systems
Speed	Fast response (1–3 seconds)	Moderate response time
Accuracy	High accuracy in functionality	Varies depending on platform
Cost	Free (academic project)	Some platforms require subscription

The proposed system provides competitive performance compared to existing systems. It offers fast response time and accurate functionality while being cost-effective, making it suitable for educational use .

9.3 Achievement of Objectives :

The main objectives defined in Chapter 1 were successfully achieved through the implementation of the proposed system.

- The system provides a project-based learning environment, which helps learners apply theoretical knowledge in practice.
- A complete platform was developed including authentication, learning paths, and project management.
- The system supports interactive learning through features such as the code editor and project execution.
- User progress tracking and dashboard functionalities were successfully implemented.

Overall, the system meets its intended goals and provides an effective solution for enhancing programming education.

9.4 Strengths and Weaknesses Analysis :

○ Strengths

- The system provides a project-based learning approach, which enhances practical skills compared to traditional learning methods.
- It integrates multiple features such as learning paths, project management, and code execution in one platform.
- The system offers a user-friendly interface, making it easy for learners to navigate and use.
- It supports interactive learning, allowing users to write and execute code directly داخل النظام.

○ Weaknesses

- Some advanced features such as AI-based recommendations and automation are not implemented.
- The system lacks performance optimization under heavy load, as it was tested under normal conditions only.
- Testing was mainly manual, without automated testing tools.
- The platform is currently web-based only and does not support mobile applications.

Despite these limitations, the system provides a solid foundation that can be improved and extended in future work.

9.6 Summary :

This chapter presented the system implementation results and evaluated the proposed platform through comparative analysis, achievement of objectives, strengths and weaknesses analysis, and KPIs. The results demonstrated that the system provides an effective and practical solution for project-based programming education.

Chapter 10 : Conclusion and Recommendations

10.1 Project Summary and Main Achievements :

This project presented a Project-Based Programming Education System designed to enhance learning by combining theoretical knowledge with practical application. The system provides structured learning paths, real-world projects, and an interactive environment that supports learners in developing their programming skills.

The main achievements of this project include the successful implementation of a web-based platform with user authentication, learning path management, project management, and an integrated code editor. Additionally, the system supports progress tracking and project evaluation, offering a more practical and engaging learning experience.

10.2 Challenges and Difficulties:

During the development of this project, several challenges were encountered:

Technical Challenges:

Some difficulties were faced while integrating the frontend (React) with the backend (Django), especially in handling APIs and data exchange. Implementing advanced features such as the code editor and execution functionality also required additional effort.

Time Constraints:

The project had to be completed within a limited academic timeframe, which required careful task prioritization and time management.

Organizational Challenges:

Managing multiple modules and features within the system required good organization and coordination to ensure consistency and smooth progress.

Despite these challenges, they were gradually resolved through continuous testing, learning, and effective planning.

10.3 Future Work:

Several improvements can be made to enhance the system in the future:

- Adding AI-based features such as personalized project recommendations and adaptive learning paths.
- Improving the code editor with more advanced features like real-time collaboration and auto-completion.
- Supporting multiple programming languages instead of a single language.
- Developing a mobile application to increase accessibility.
- Enhancing the evaluation system with automated code analysis and feedback.

10.4 Recommendations:

Based on the development and evaluation of this project, the following recommendations are suggested:

- It is recommended to adopt project-based learning approaches in programming education to improve practical skills.
- Institutions can benefit from integrating similar systems to enhance student engagement and learning outcomes.
- Future developers should focus on improving system scalability and adding advanced features.
- Continuous testing and user feedback should be considered to ensure system improvement and usability.

10.5 Summary :

This chapter concluded the project by summarizing the main achievements, discussing the encountered challenges, and presenting recommendations and future improvements. The proposed system successfully achieved its educational objectives and provides a strong foundation for future development and enhancement.